

Design Techniques for Wideband Single-Bit Continuous-Time Delta-Sigma Modulators with FIR Feedback DACs

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Daniel Tyukov, 5714699 - ET4278

Q1. Introduction

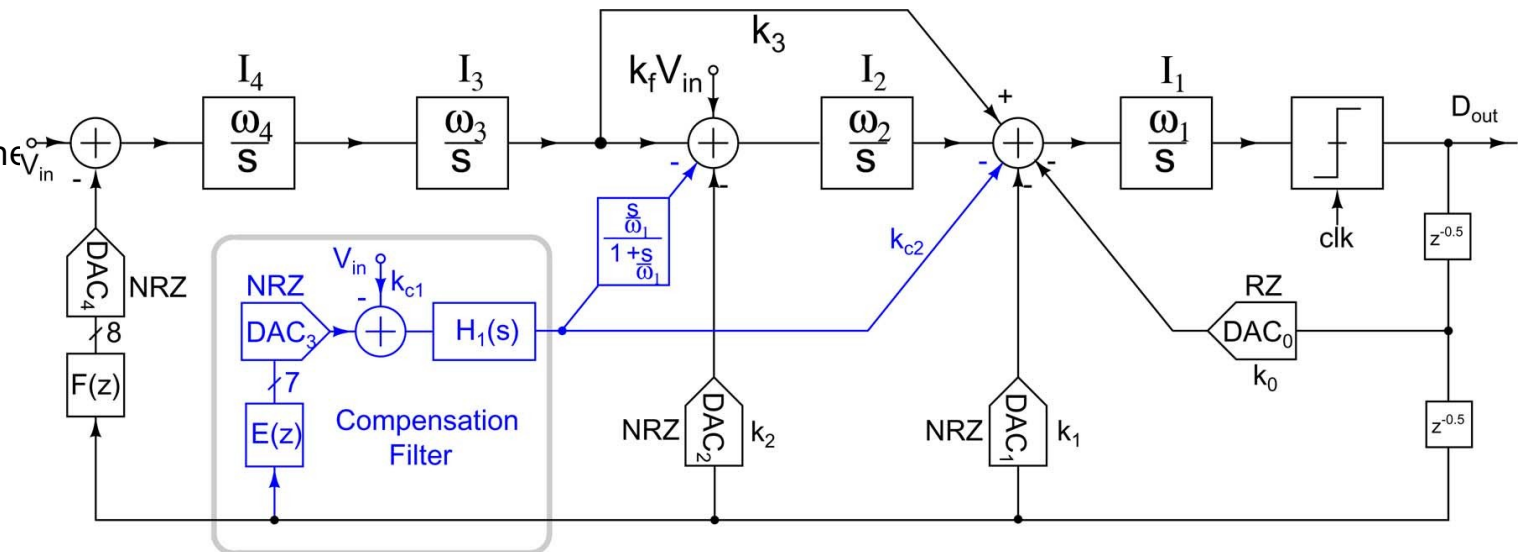
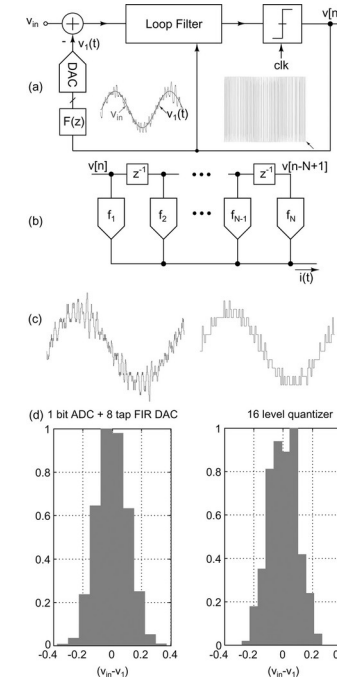
- **Issues addressed by the paper**

- Single-bit NRZ feedback: full-scale steps every cycle -> high clock-jitter sensitivity
- Comparator metastability adds data-dependent jitter -> in-band SNR collapses
- Full-scale single-bit feedback demands a very linear loop filter
- Multi-bit avoids these but costs quantizer power/area + DAC mismatch -> DEM/calibration
- An FIR feedback DAC adds group delay (excess loop delay) that destabilises the loop

- **Main techniques**

- 1-bit ADC + 8-tap semi-digital FIR feedback DAC (multi-bit-like waveform, 2-level linear, no DEM)
- First FIR tap = 0 (explicit 1-cycle delay) to dodge metastability
- Analog FIR-delay compensation: $E(z)$ FIR DAC + bandpass $H_1(s)$ restores the target NTF
- RZ DAC0 for ELD compensation (no summer around the quantizer)
- Modified CIFB + input feed-forward, active-RC integrators, assisted opamp

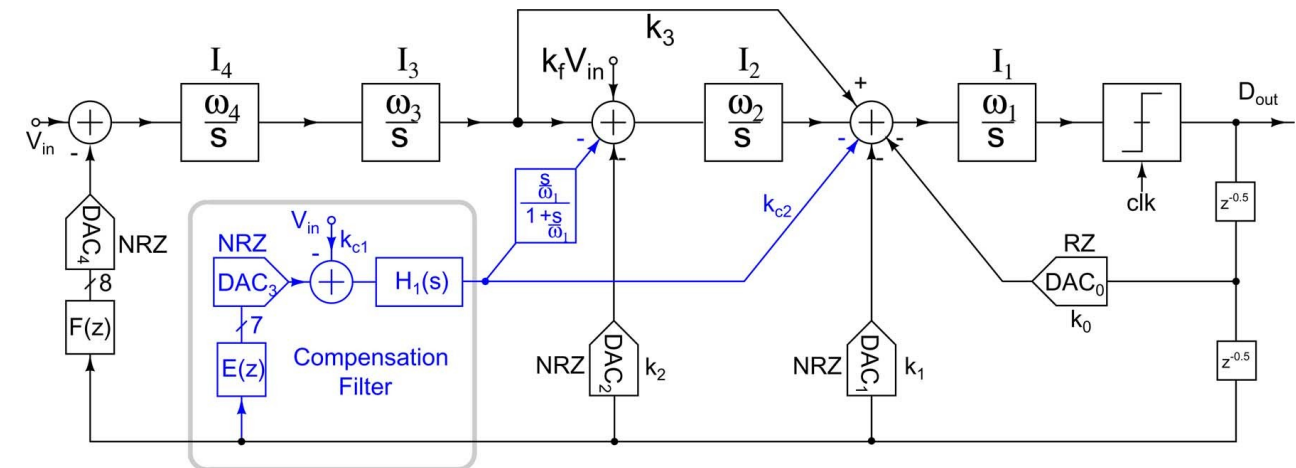
- **Conclusion: FIR feedback gives a 1-bit loop the jitter/linearity of a multi-bit loop; the price is loop delay, solved by an analog compensation path.**



Q2. Model and coefficients

- **Coefficients**

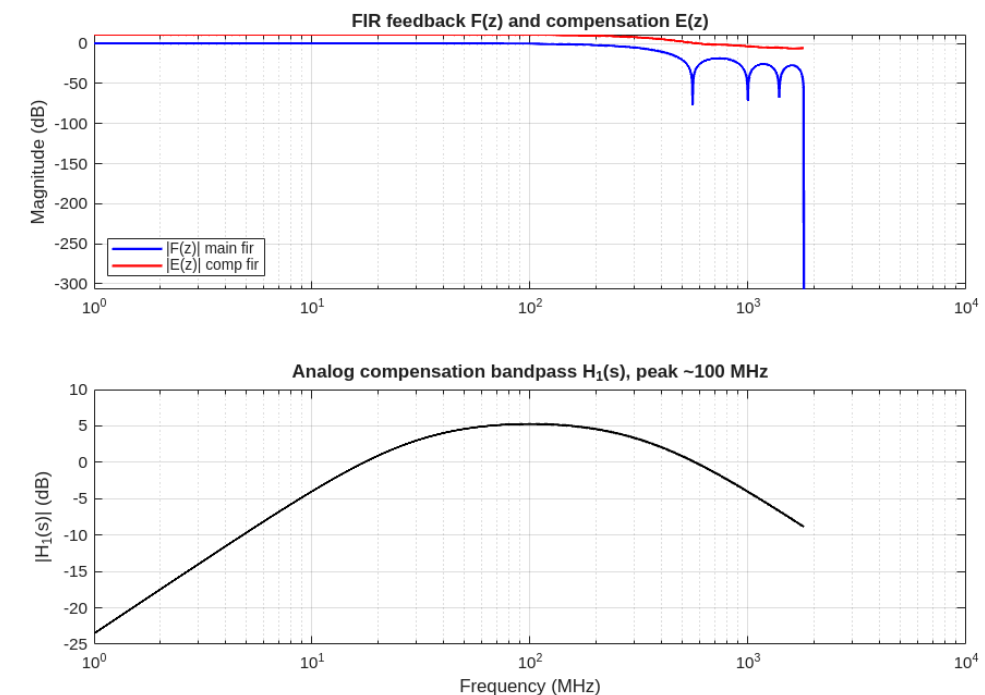
- ⊖ F(z) main FIR, 8 taps: [0.07 0.12 0.15 0.16 0.16 0.15 0.12 0.07], sum = 1
- ⊖ E(z) compensation FIR, 7 taps (derived): [0.93 0.81 0.66 0.50 0.34 0.19 0.07]
- ⊖ CIFB feedback gains a = [0.0060 0.0457 0.1933 0.5547]
- ⊖ Resonator gains g1 = 8.35e-4, g2 = 3.13e-3 (NTF zeros at 0.46, 0.89 of fb)
- ⊖ Input feed-forward kff = 1.2 into last integrator (STF peaking); H1(s) bandpass ~100 MHz
- ⊖ fs = 3.6 GS/s, fb = 36 MHz, OSR = 50, order = 4, OBG = 1.5



- **Derivation**

- ⊖ Synthesise 4th-order OBG-1.5 NTF (Butterworth-HP poles bisected for OBG) + 2 optimised zeros
- ⊖ Match the loop characteristic polynomial $\rightarrow$ CIFB coefficients a, g
- ⊖ $F(z)$ from Eq.(1); $E(z) = (1 - F(z))/(1 - z^{-1})$ (DC gain 0)
- ⊖ Fit $H1(s)$ to the replica path by impulse invariance

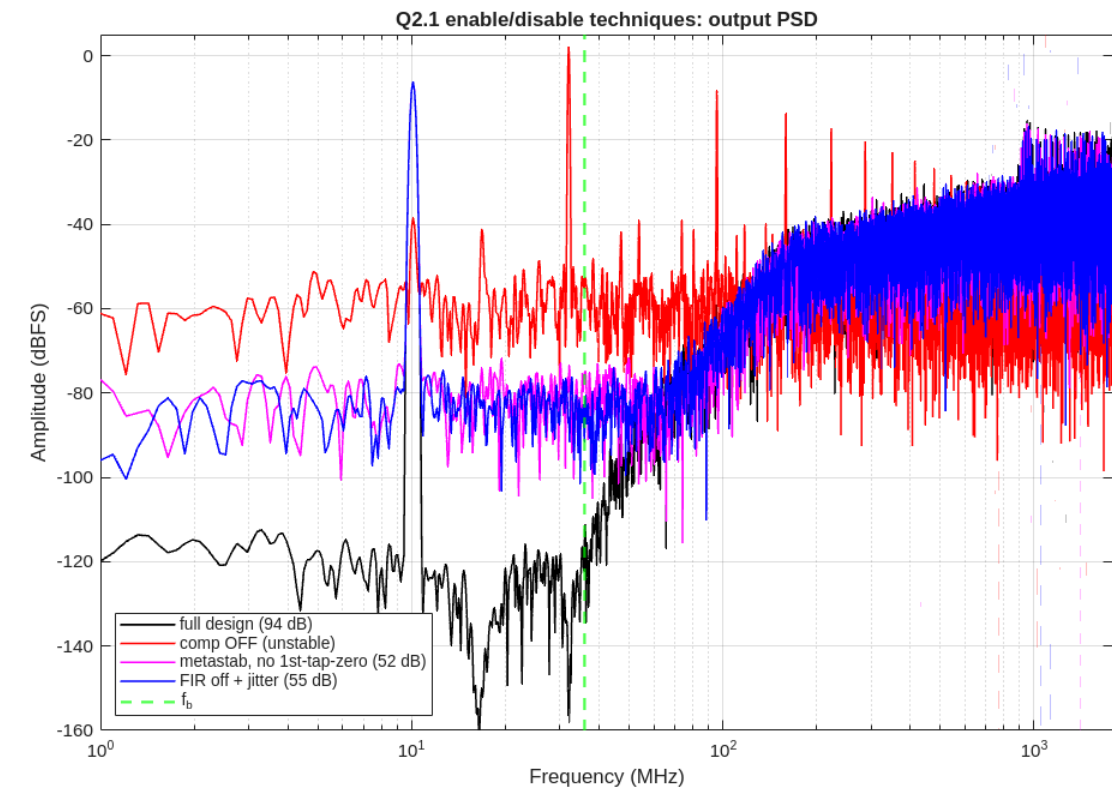
- **Conclusion:** $F(z)$ is taken from the paper; $E(z)$, a , g and $H1(s)$ were derived. The model reproduces the OBG-1.5 NTF (Fig 8).



Q2.1 Techniques on/off

- Each technique enabled then disabled, quantified in-band (36 MHz):

Technique	OFF	ON
Compensation path	unstable (~ -45 dB)	~ 94 dB
First-tap = 0 (metastability)	~ 52 dB	~ 91 dB
FIR DAC (jitter floor)	plain 1-bit	~ 25 dB lower
Ideal SQNR (all on)	-	~ 94 dB



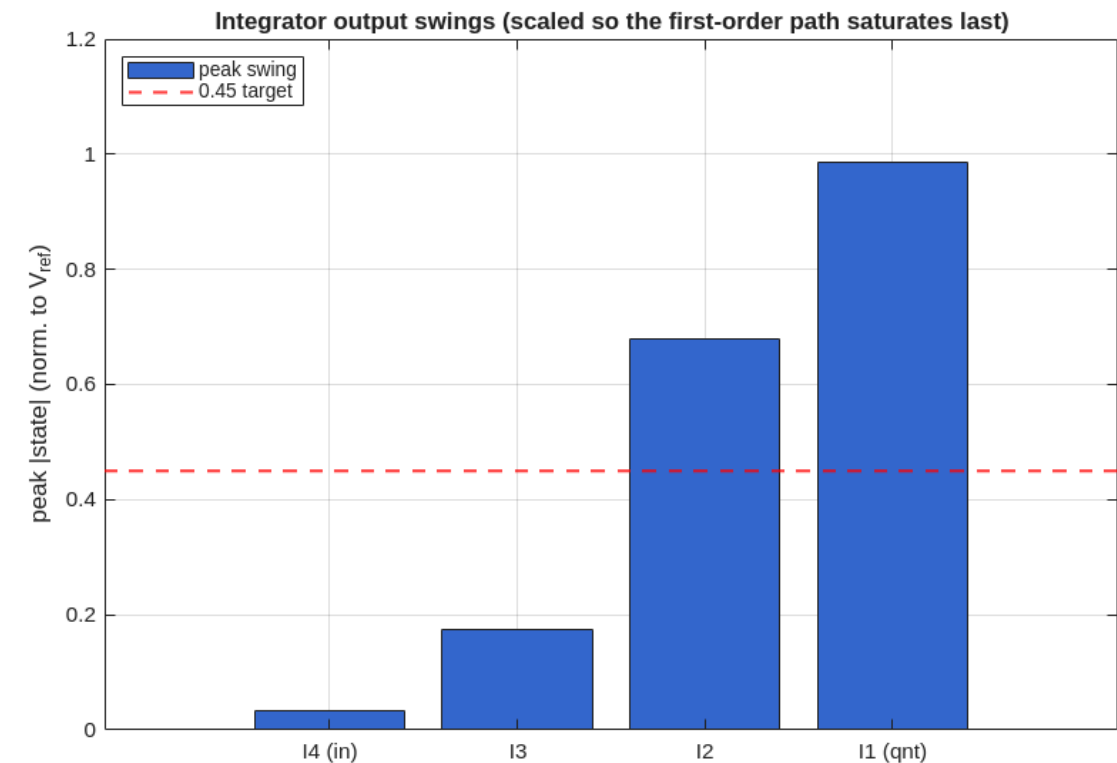
Conclusion: compensation = stability, first-tap-zero = +39 dB (metastability), FIR DAC = ~ 25 dB jitter-floor reduction.

Q2.2 Voltage scaling

- **Assumptions**

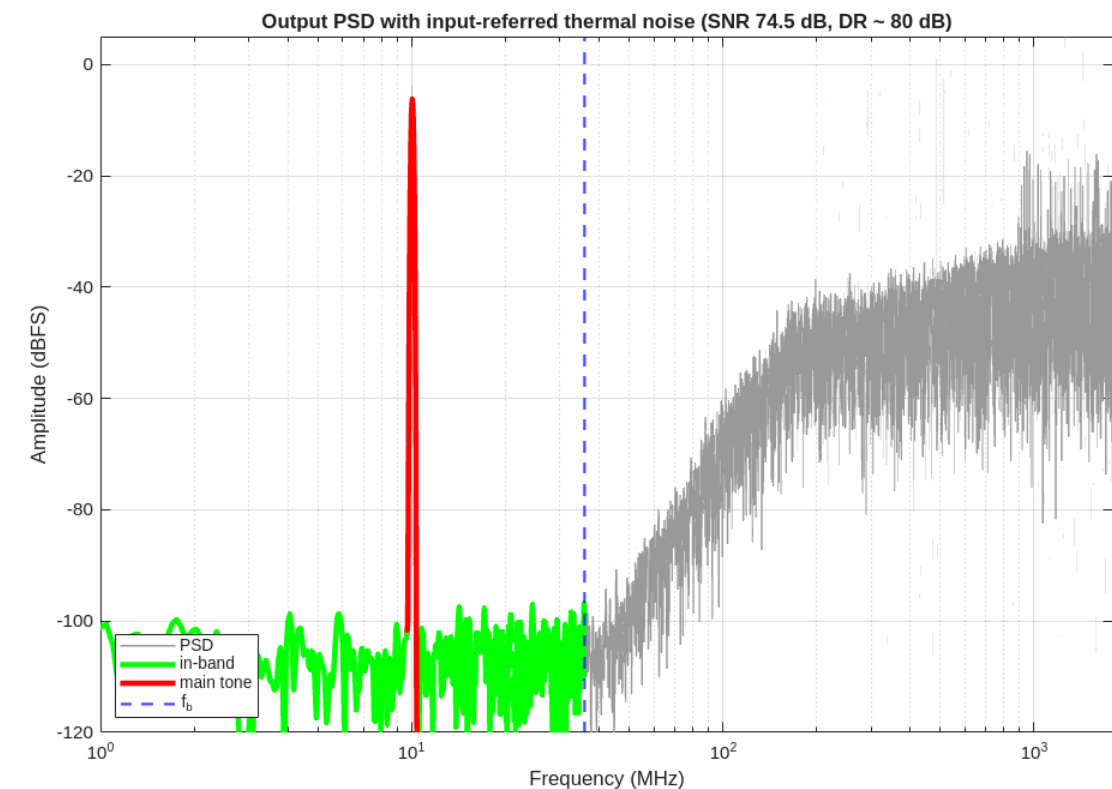
- ⊖ Internal full scale = ± 1 ; quantizer output in $\{-1, +1\}$
- ⊖ Integrator output swings scaled to stay below ~ 0.45 (first-order path saturates last)
- ⊖ States map to the 1.2 V supply / V_{ref} when expressed in volts
- ⊖ Scaling redistributes gain between integrators; the loop transfer is unchanged

- **Conclusion: scaling preserves the NTF/STF (1-bit modulator) and keeps every integrator swing inside the MSA.**



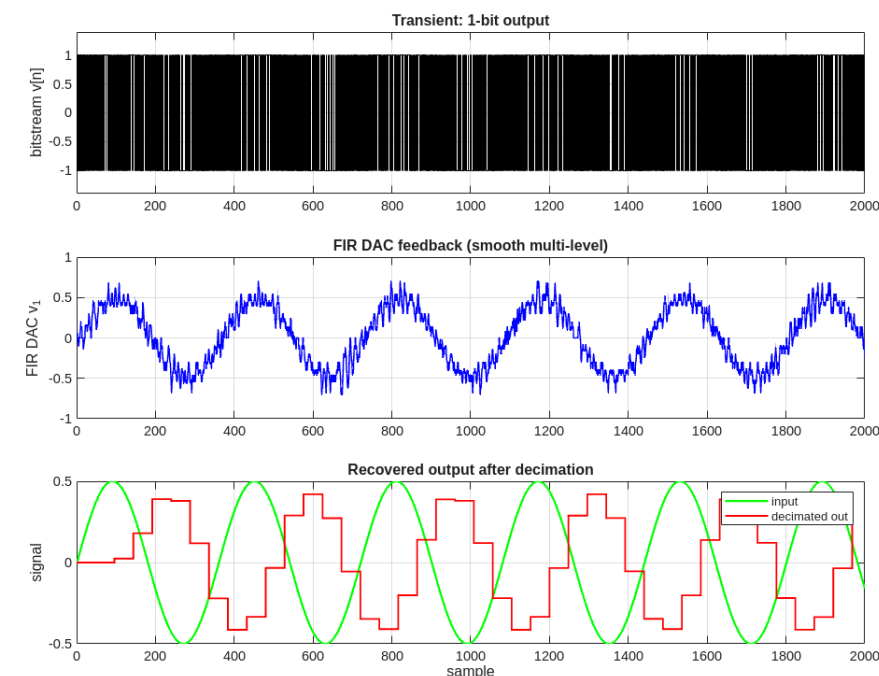
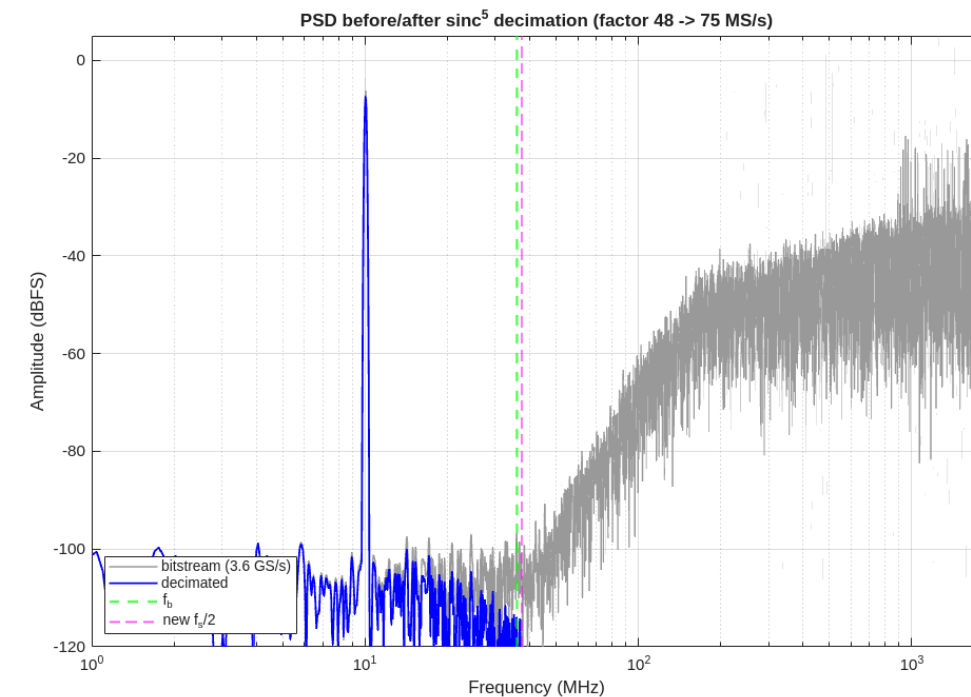
Q2.3 Thermal noise & spectrum

- **Thermal noise**
 - ⊖ Input-referred Gaussian noise added at the I4 summing node
 - ⊖ sigma sized so the in-band floor gives DR ~ 83 dB at 36 MHz (matches Table I)
 - ⊖ Noise is unshaped (flat): it sets the floor, separate from the shaped quantisation noise
 - ⊖ Output PSD scaled in dBFS by $A \cdot N/2$ so it overlays the paper
- **Conclusion:** with the chosen input-referred noise the modulator reaches DR ~ 83 dB over 36 MHz.



Q2.4 Decimation filter

- **Decimation filter**
 - ⊖ $\text{sinc}^{(\text{order}+1)} = \text{sinc}^5$ CIC, implemented as cascaded box-car averages
 - ⊖ Decimation factor $\sim$ OSR (50); downsample to $\sim$ Nyquist
 - ⊖ Plot output spectrum before/after decimation + transient waveforms
 - ⊖ (bitstream, integrator states, recovered sine)
- **Conclusion: decimation removes the out-of-band shaped noise; the in-band SNR is preserved.**



Q2. Simulate the SD architecture described in the paper using a Matlab script.

Q2.5 List of simulation results

- Paper results and which were reproduced in MATLAB:

Result	Paper value	Repro
NTF (Fig 8)	4th-order, OBG 1.5, notch ~ -90 dB	yes
STF (Fig 9)	+12 dB peak ~100 MHz	yes
Output PSD (Fig 22)	SNDR 70.3, SNR 74.9, HD2 73.5 dB	yes
SNR/SNDR vs amp (Fig 21)	pk SNR 76.4, SNDR 70.9, DR 83	yes
Jitter PSD (Fig 2)	FIR ~ 4-bit, << plain 1-bit	yes
Jitter nulls (Fig 26)	minima at F(z) zeros ~600/800 MHz	yes
Jitter tolerance (Fig 27)	~10x vs plain 1-bit	yes
Metastability (Fig 4)	52 -> 91 dB	yes
Nonlinear integrator (Fig 3)	floor rise = 4-bit case	yes
ISI correction (Figs 29/30)	HD2 -15 dB, SNDR -> 74.3	yes
Survey scatter (Figs 23/24)	BW/DR & P/SNDR clouds	no (survey)

MATLAB assumptions

fs = 3.6 GS/s, OSR = 50

Nfft = 2¹⁶

window kaiser(N, 20)

coherent input tone

ns = 20 CT sub-steps/clock

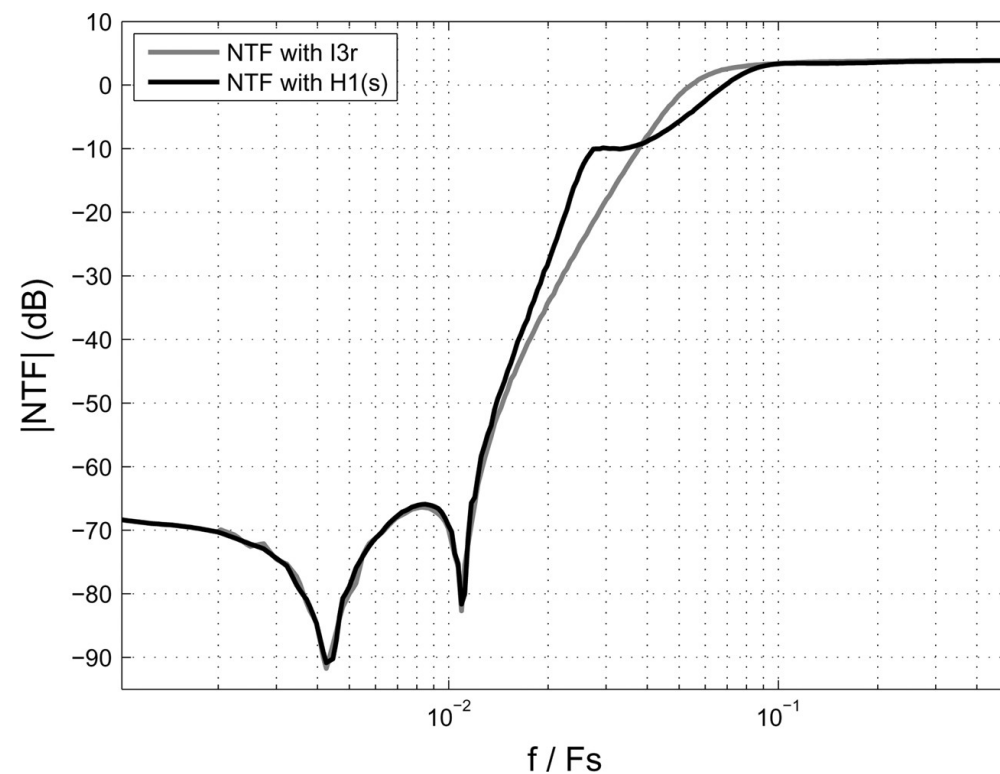
rng(1) seeded

Conclusion: reproduced NTF, STF, PSD, SNR/SNDR-vs-A, jitter (2/26/27), metastability (4), nonlinear integrator (3), ISI (29/30); survey scatter (23/24) not reproducible.

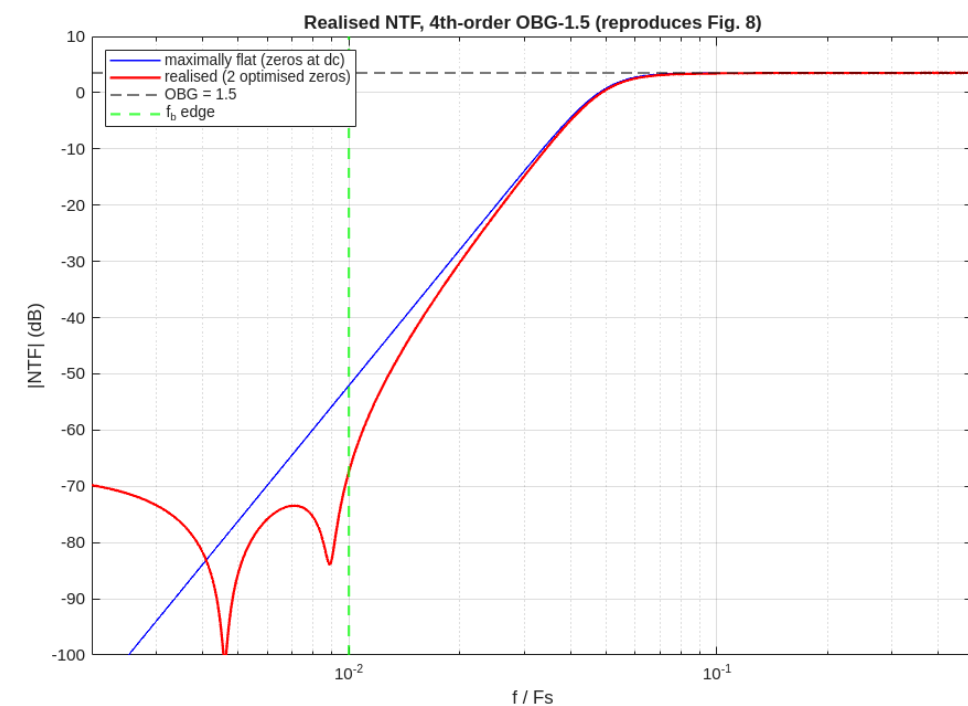
Q2.6A NTF

- Simulated: Realised $|NTF|$ vs f/F_s : 4th-order, OBG = 1.5, two optimised in-band notches.

Published result - FIG08



My result - MATLAB

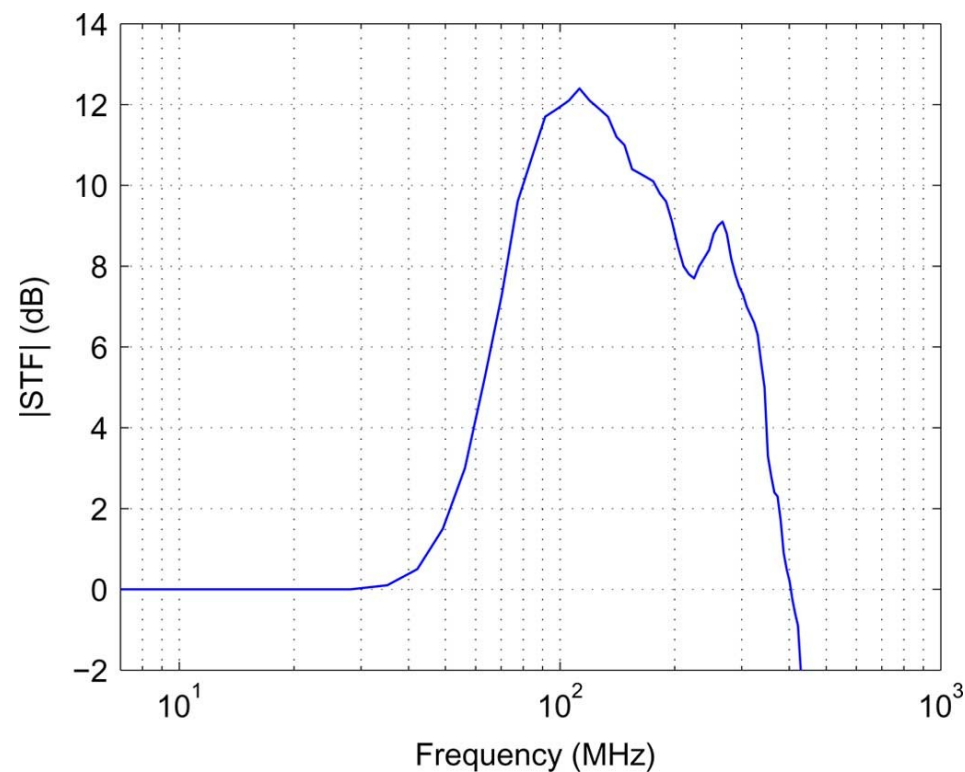


- Conclusion: Same 4th-order high-pass shape as Fig 8: OBG = 1.5 plateau, two in-band notches (~ -100 dB).

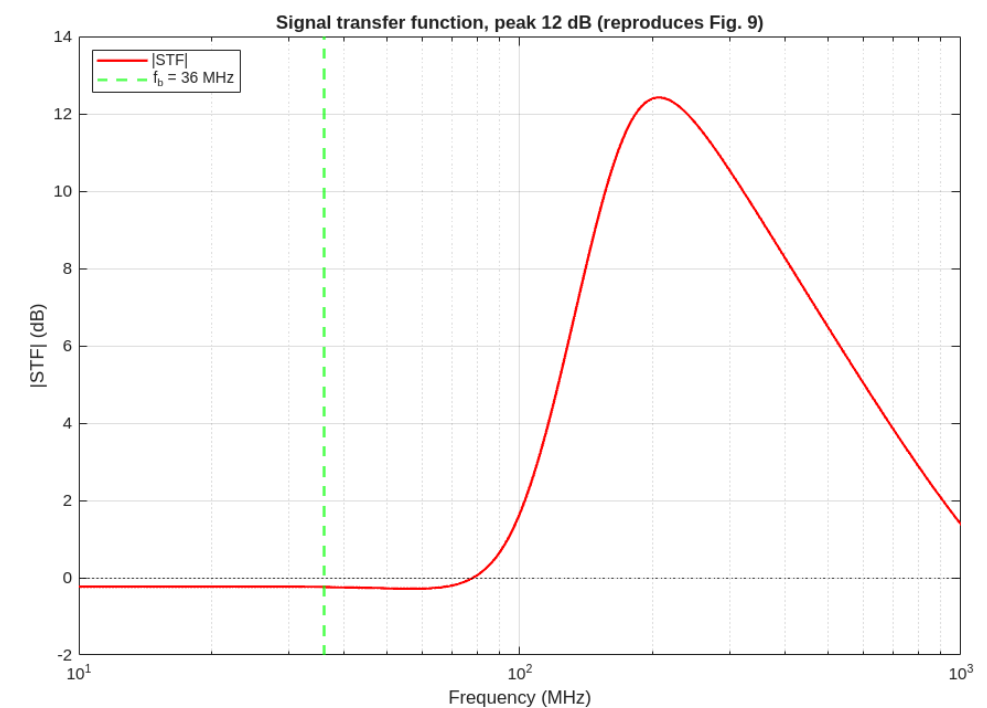
Q2.6B STF

- Simulated: $|STF|$ magnitude from the input feed-forward.

Published result - FIG09



My result - MATLAB

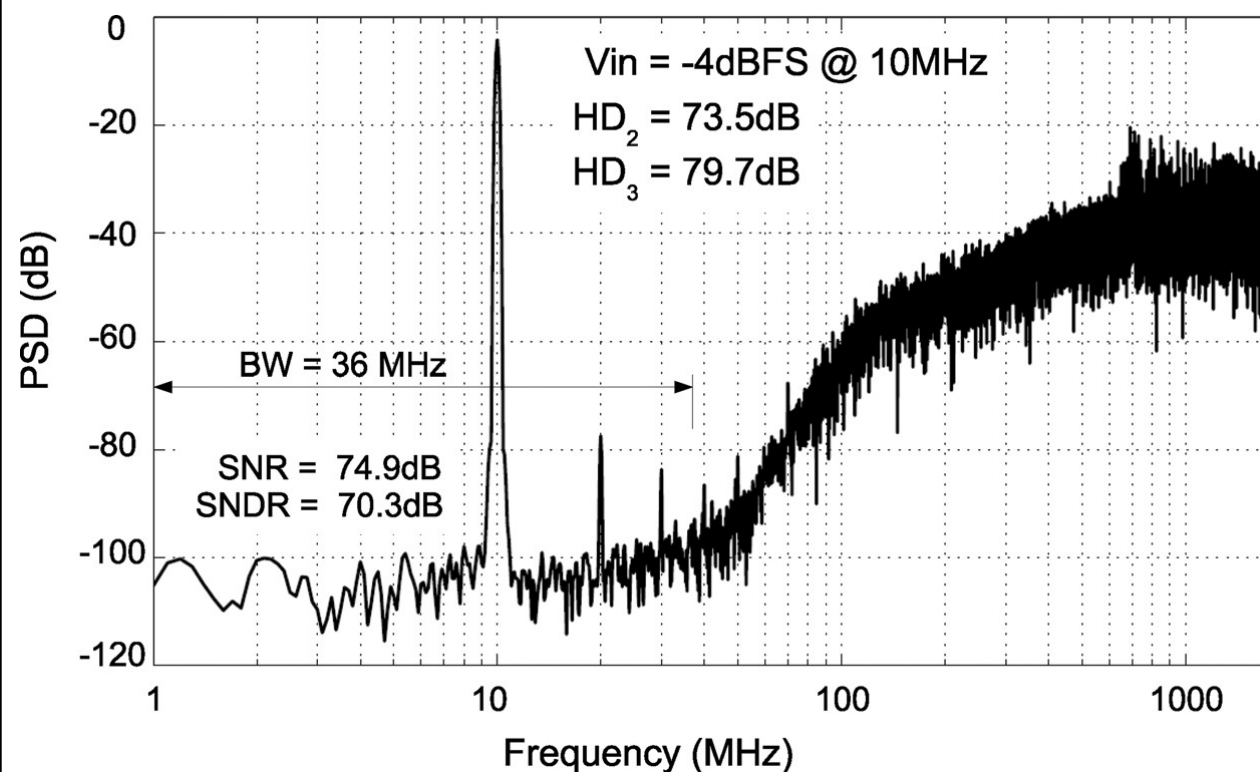


- Conclusion: Same bandpass shape as Fig 9: $\sim +12$ dB peak then rolls off (peak ~ 200 MHz vs ~ 100 MHz); needs input pre-filter.**

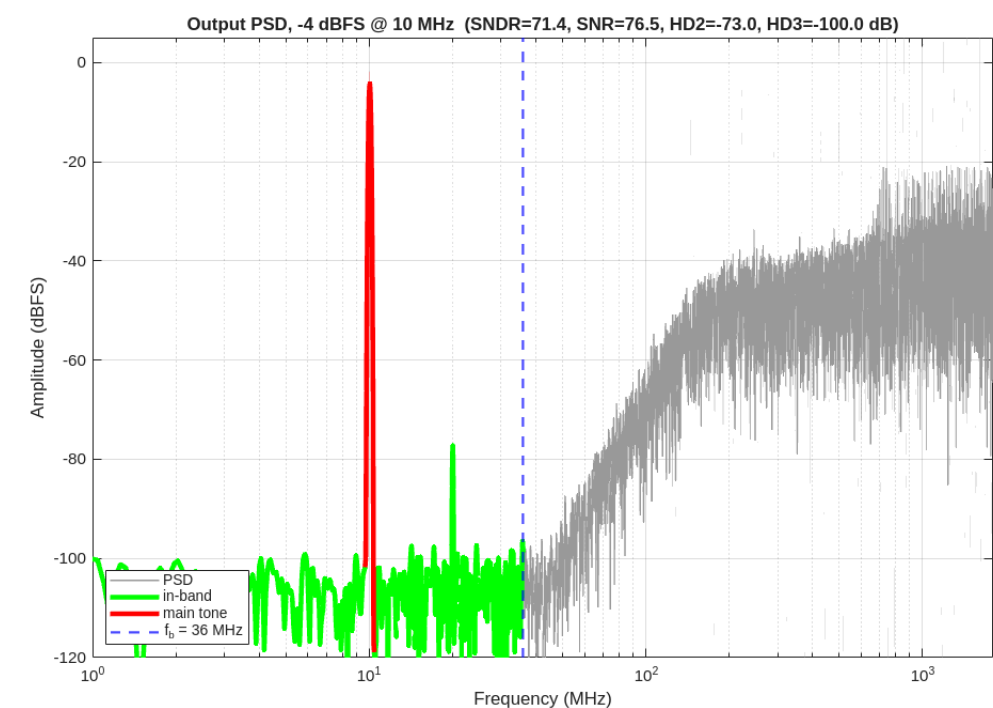
Q2.6C Output PSD

- Simulated: Output PSD, -4 dBFS 10 MHz tone, 2¹⁶-pt FFT.

Published result - FIG22



My result - MATLAB

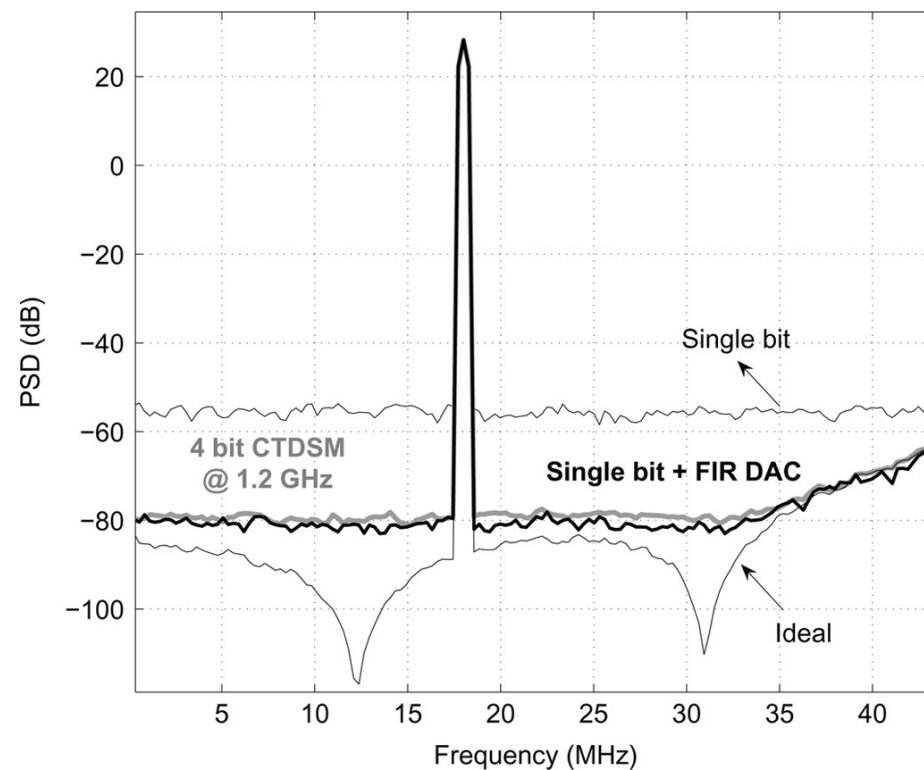


- Conclusion: SNDR 71.4 (paper 70.3), SNR 76.5 (74.9), HD2 -73.0 (-73.5) dB.

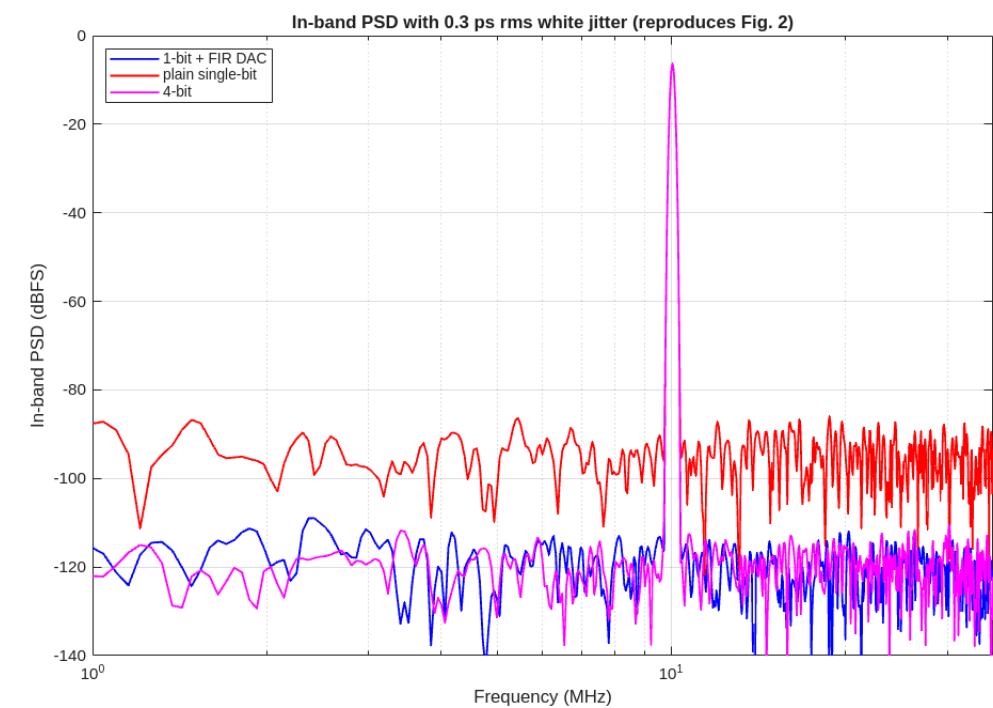
Q2.6D Jitter PSD

- Simulated: In-band PSD with 0.3 ps rms white clock jitter: FIR vs plain 1-bit vs 4-bit.

Published result - FIG02



My result - MATLAB

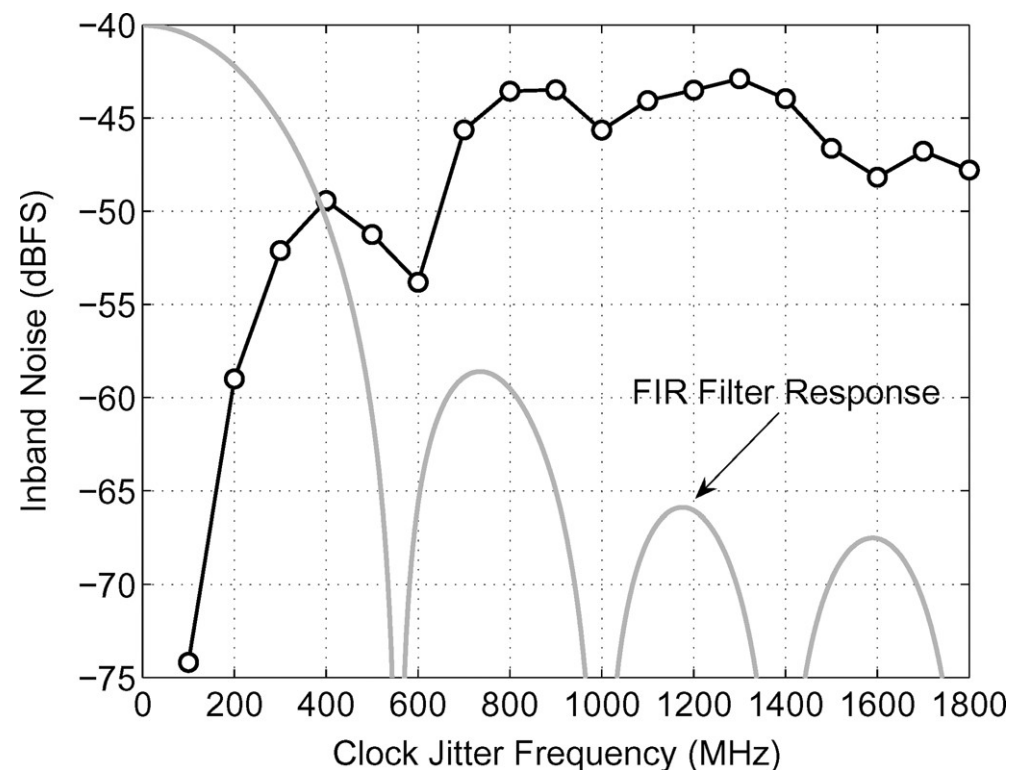


- Conclusion: FIR jitter floor ~25 dB below plain 1-bit, similar to 4-bit.**

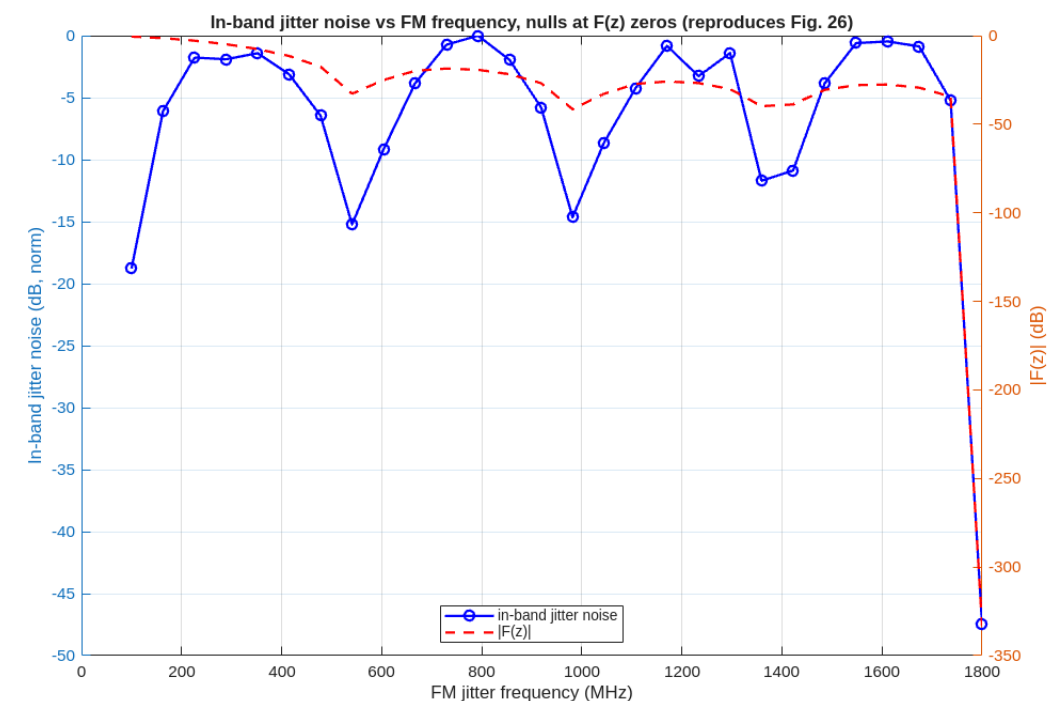
Q2.6E Jitter filtering (FM nulls)

- Simulated: In-band jitter noise vs FM jitter frequency, overlaid with $|F(z)|$.

Published result - FIG26



My result - MATLAB

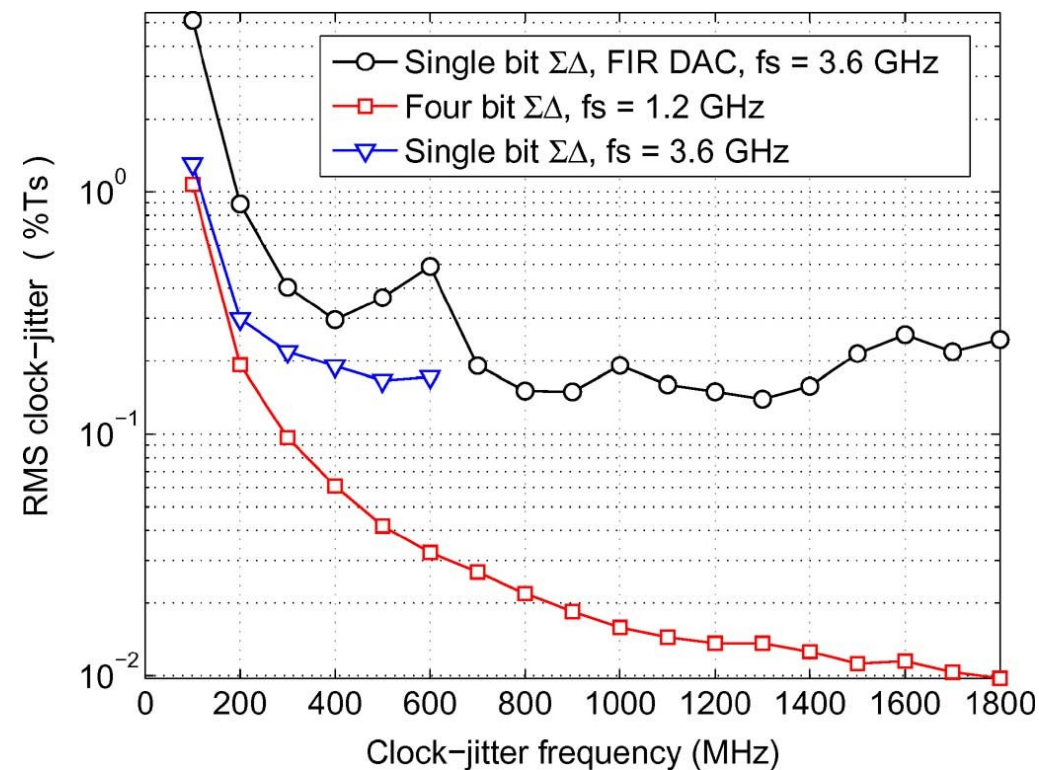


- Conclusion: In-band jitter noise nulls fall at the zeros of $F(z)$.**

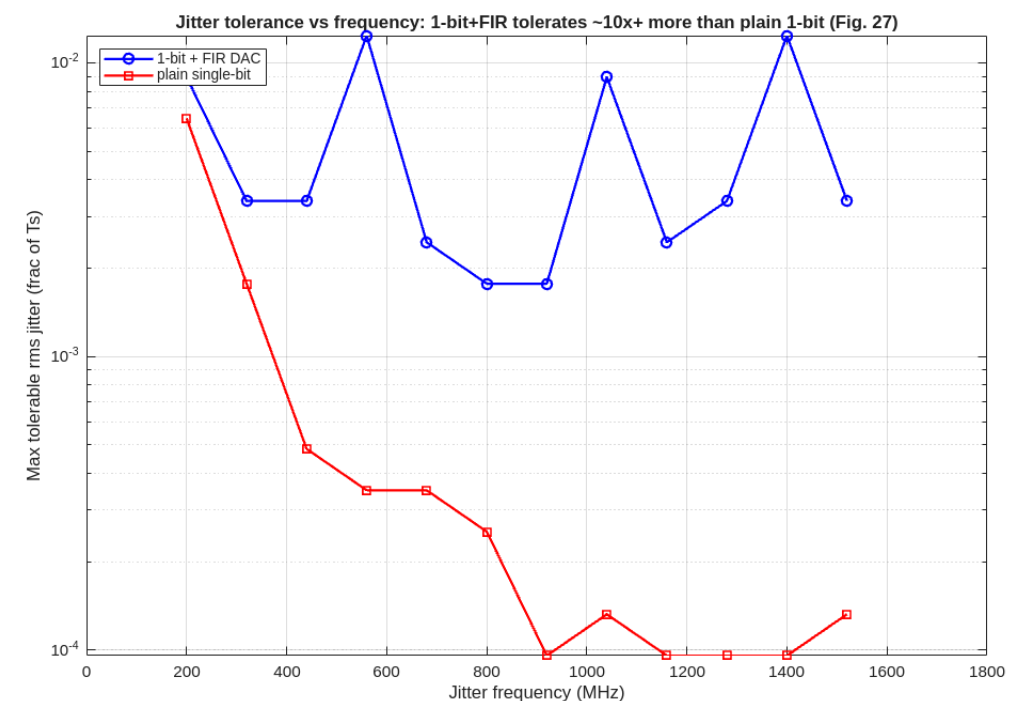
Q2.6F Jitter tolerance

- Simulated: Max tolerable rms jitter vs jitter frequency: 1-bit+FIR vs 4-bit vs plain 1-bit.

Published result - FIG27



My result - MATLAB

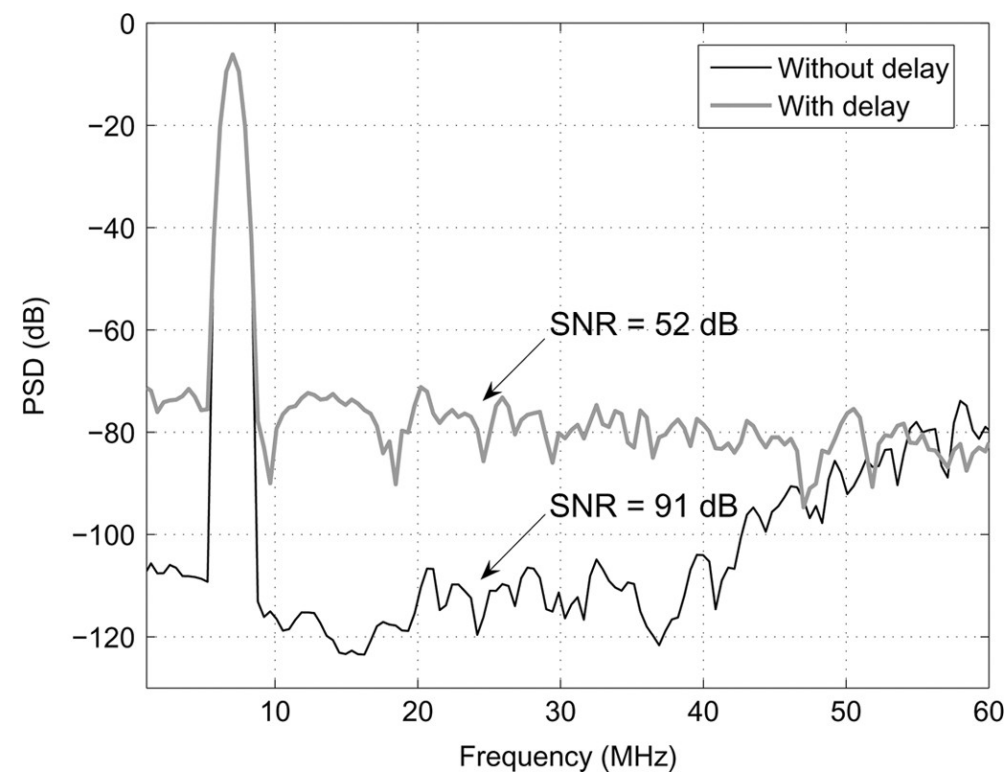


- Conclusion: 1-bit+FIR tolerates ~10x more rms jitter than plain 1-bit.**

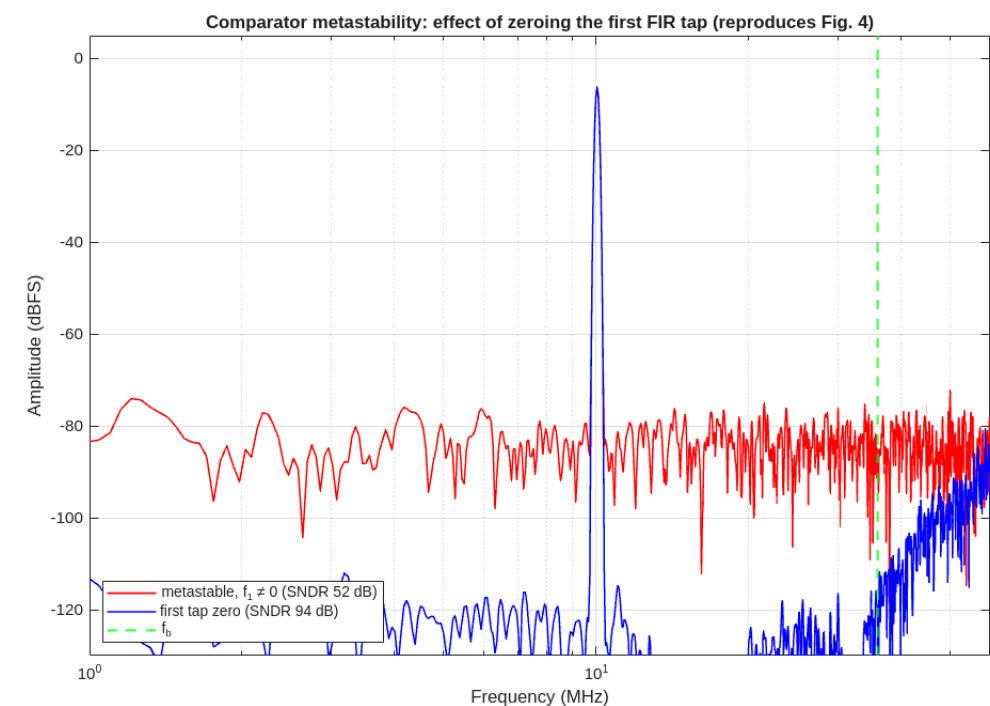
Q2.6G Metastability

- Simulated: In-band PSD with vs without the first-tap explicit delay.

Published result - FIG04



My result - MATLAB

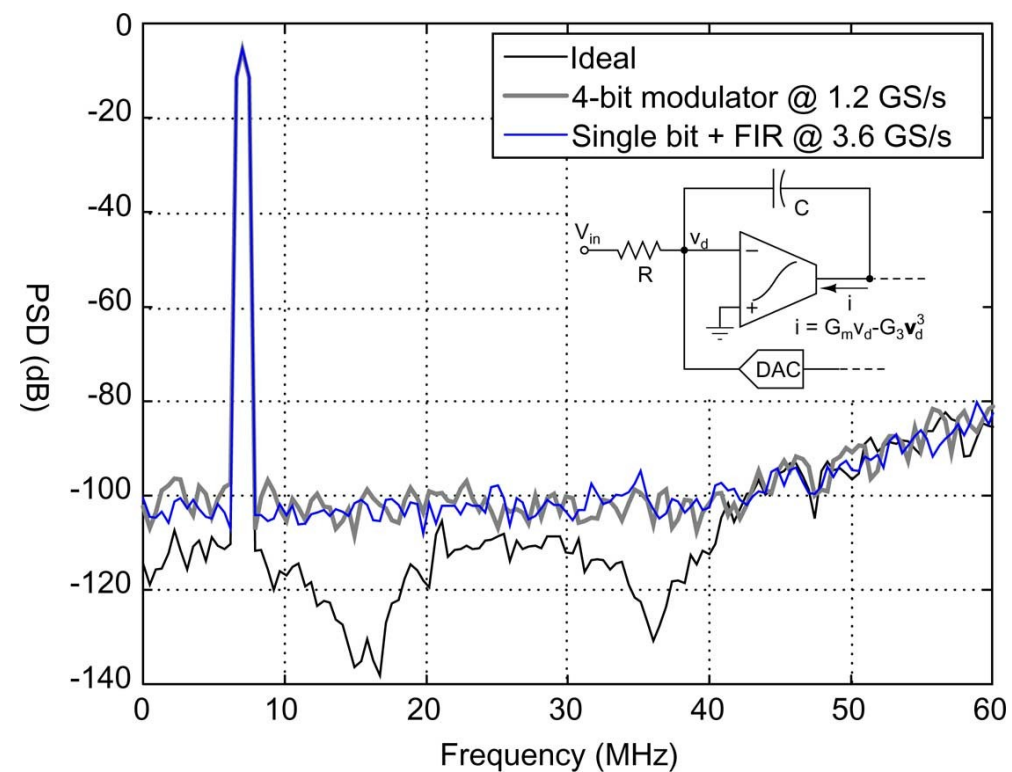


- Conclusion: First-tap-zero recovers ~52 -> ~91 dB.**

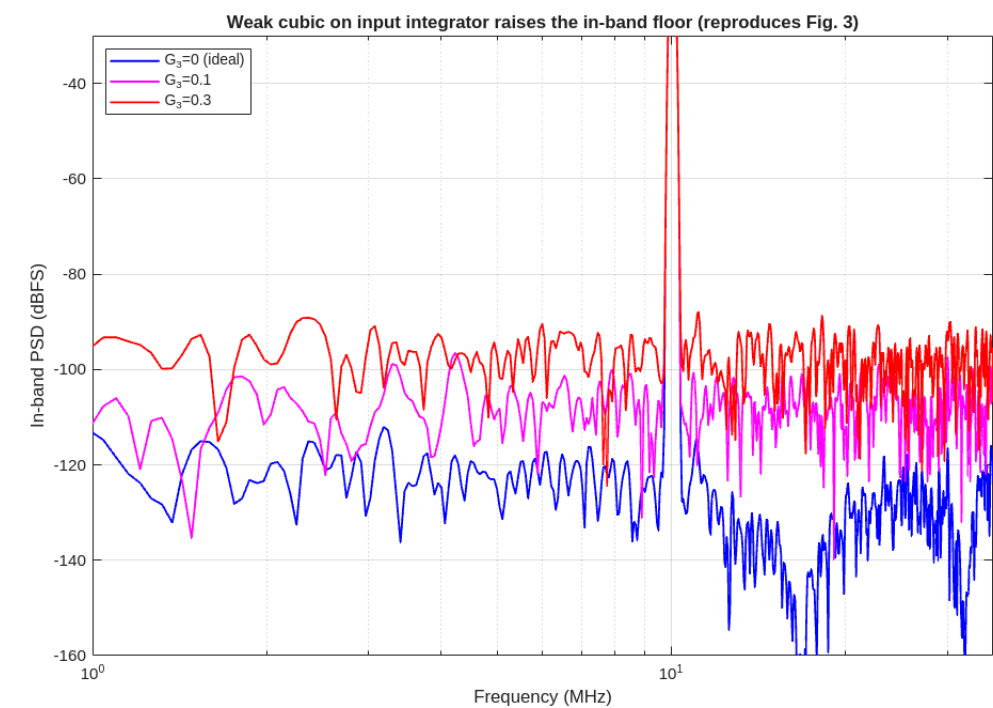
Q2.6H Nonlinear integrator

- Simulated: In-band PSD with a weak cubic on the input integrator I4.

Published result - FIG03



My result - MATLAB

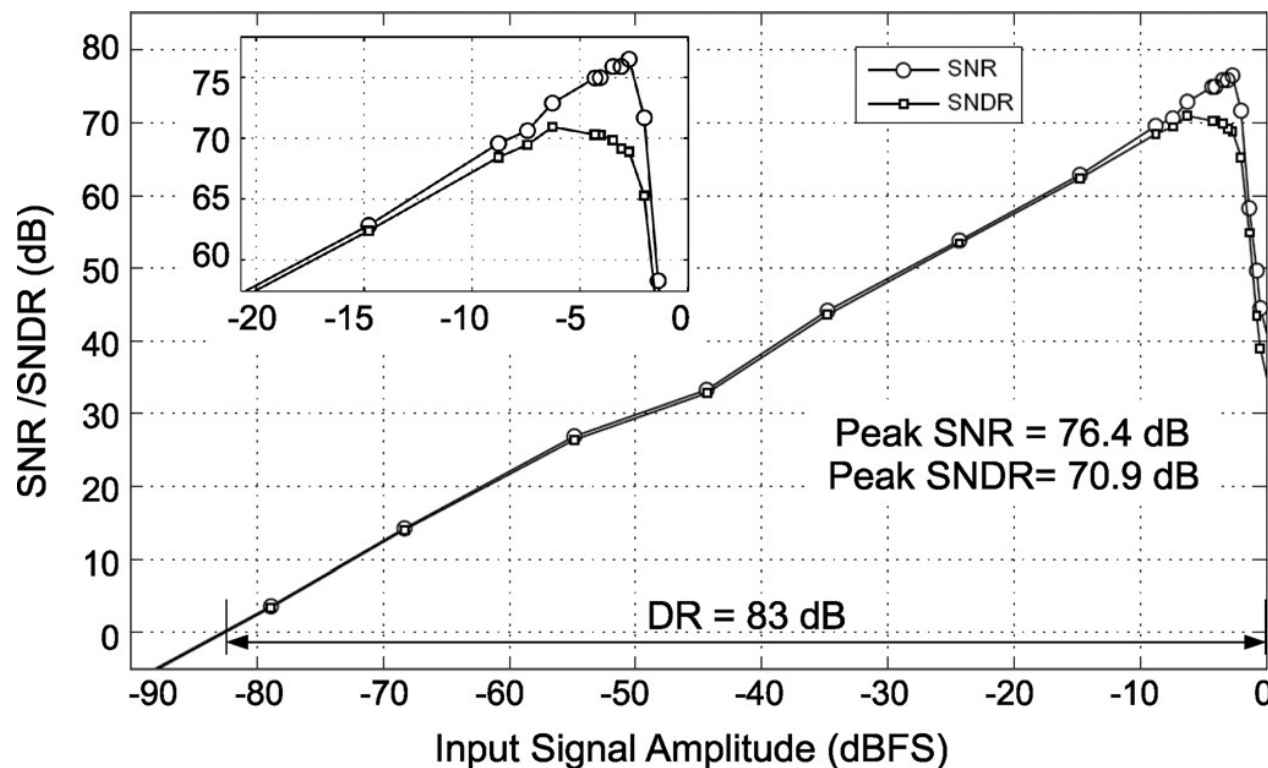


- Conclusion: Weak cubic on I4 raises the in-band floor (out-of-band noise demodulation).**

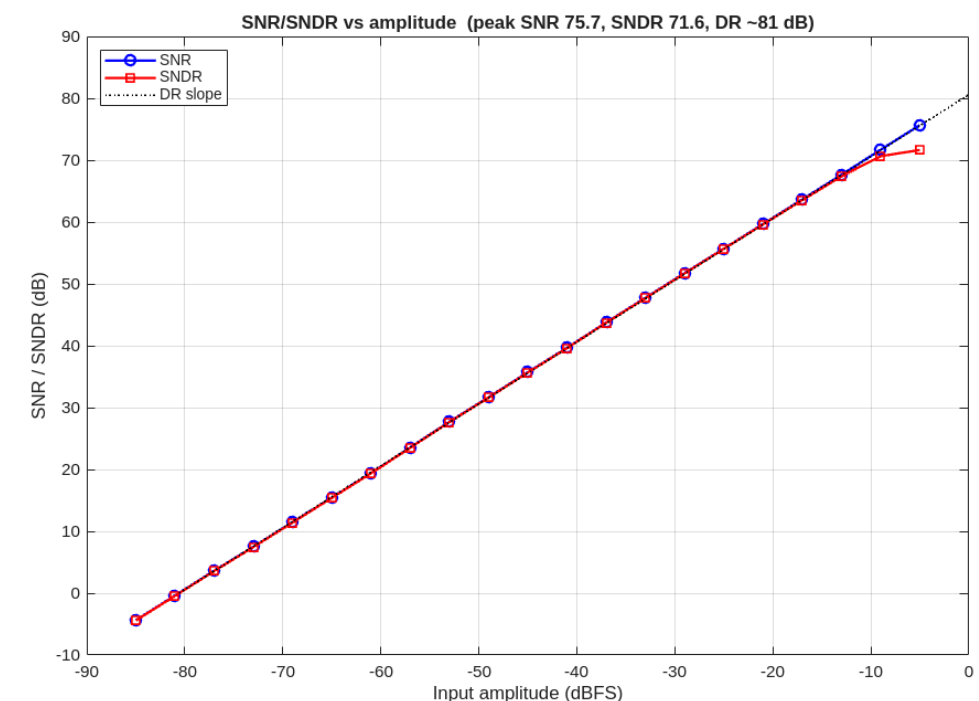
Q2.6I SNDR vs amplitude

- Simulated: SNR/SNDR vs input amplitude sweep including the ISI kink.

Published result - FIG21

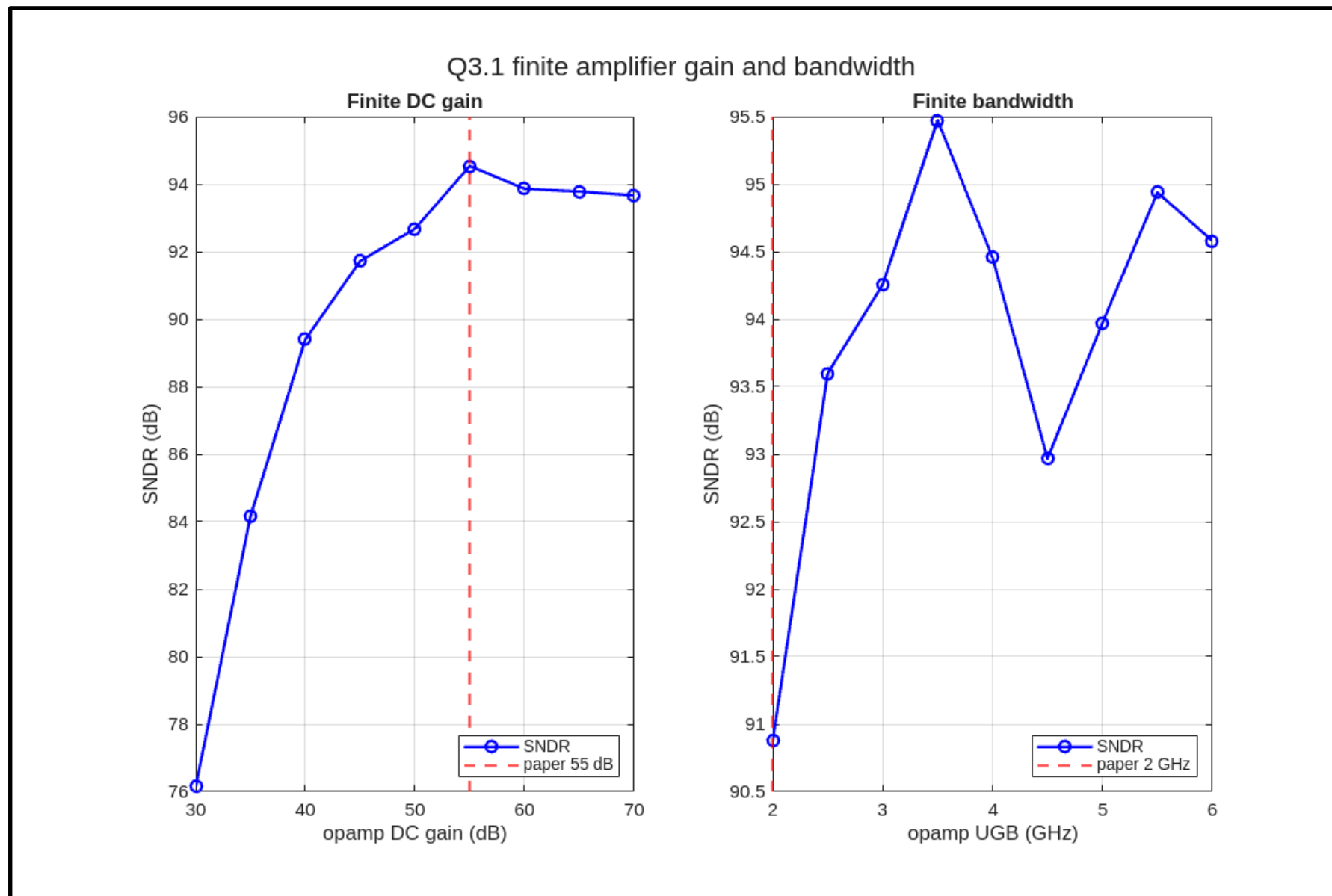


My result - MATLAB



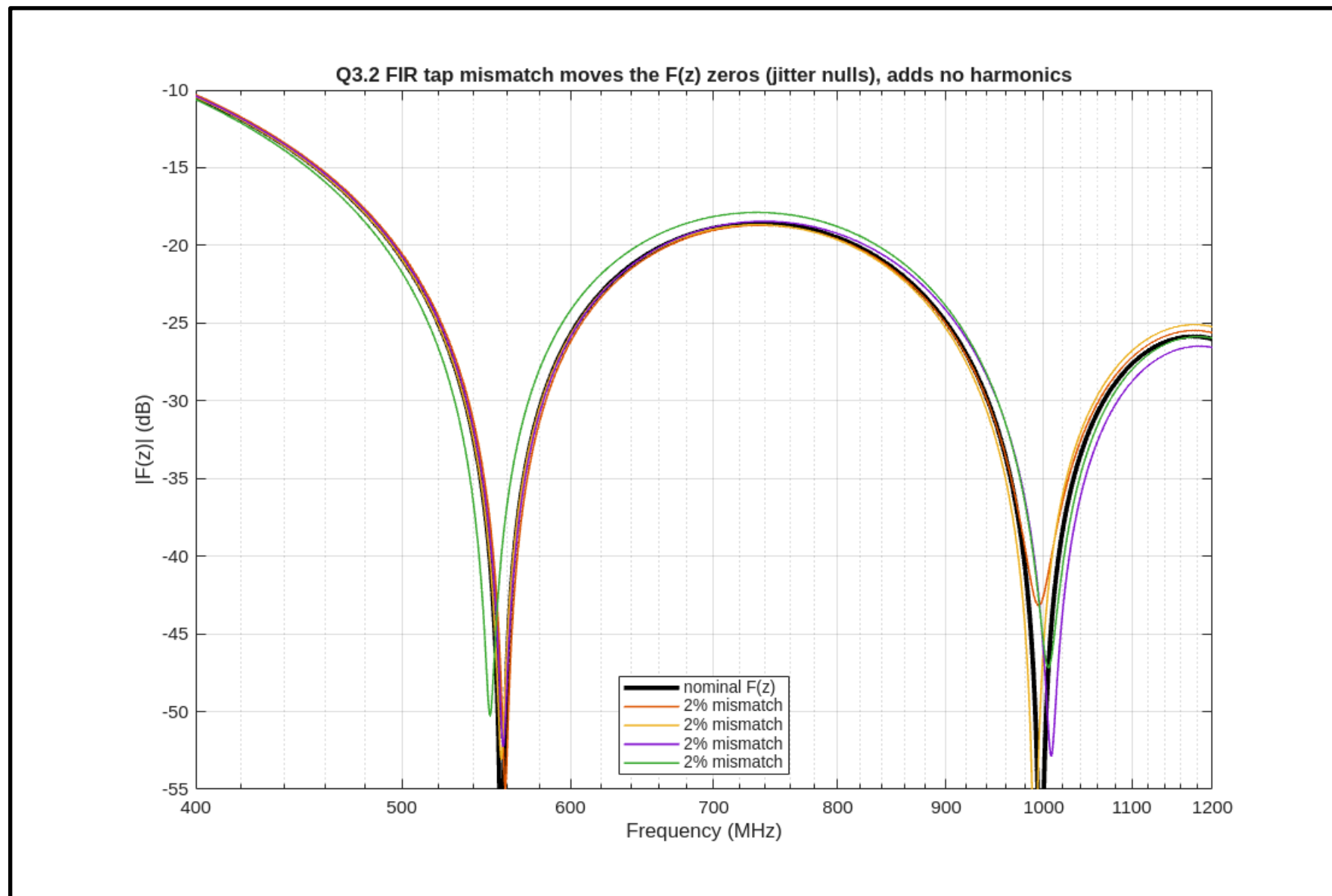
- Conclusion: Peak SNR 75.7, SNDR 71.6, DR ~81 dB; ISI kink near full scale.

Q3.1 Finite amplifier gain & BW



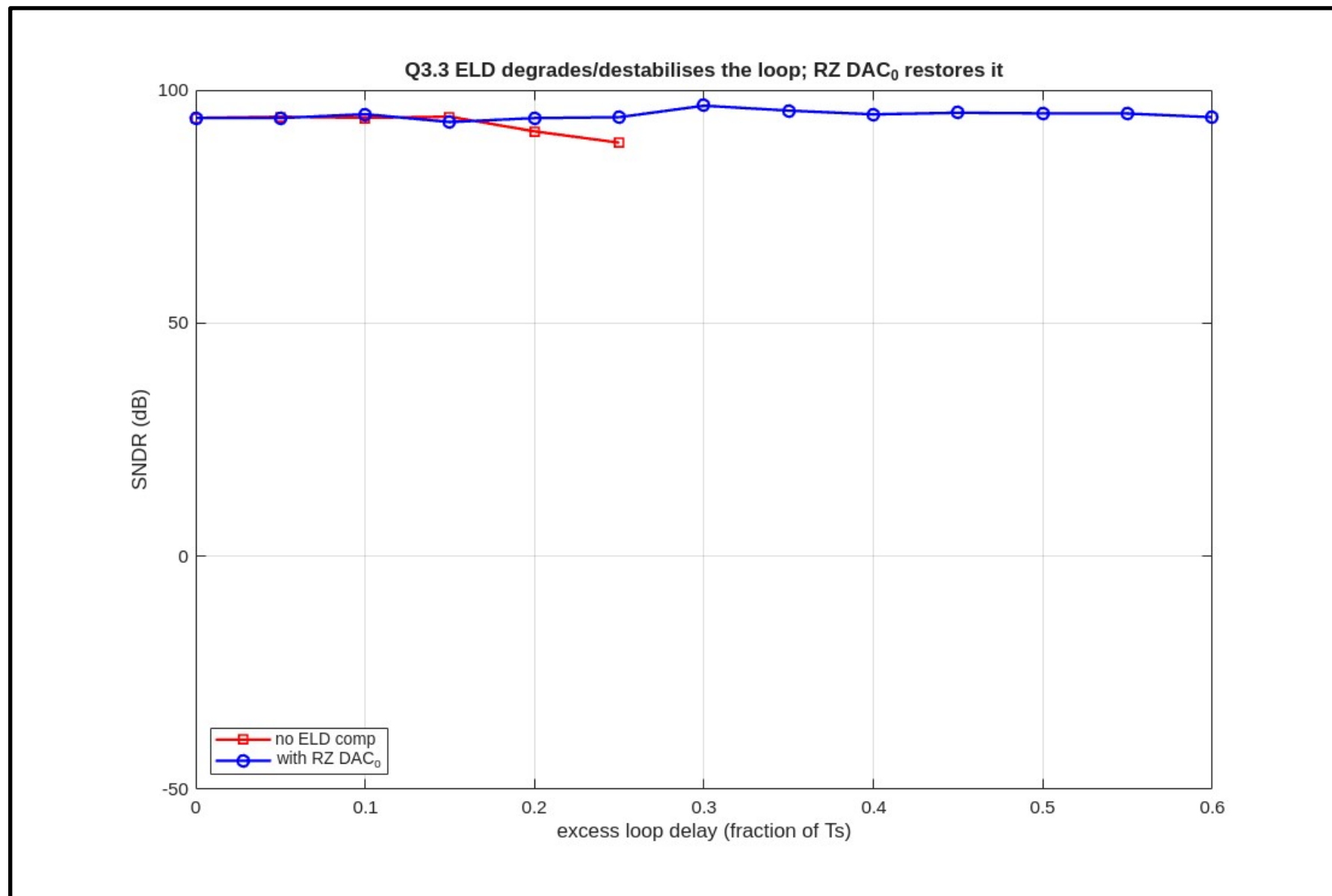
- Paper: opamp DC gain ~ 55 dB, UGB ~ 2 GHz (Fig 12).
Model: leaky integrator (finite DC gain) + 1-pole roll-off at UGB on each integrator; swept A_{dc} and UGB.
Impact: finite gain leaks the NTF notch (in-band floor rises); finite UGB adds excess phase $\rightarrow$ NTF droop.
Conclusion: negligible penalty needs ~50 dB DC gain and ~2-4 GHz UGB; the paper's 55 dB / 2 GHz is sufficient.

Q3.2 DAC mismatch



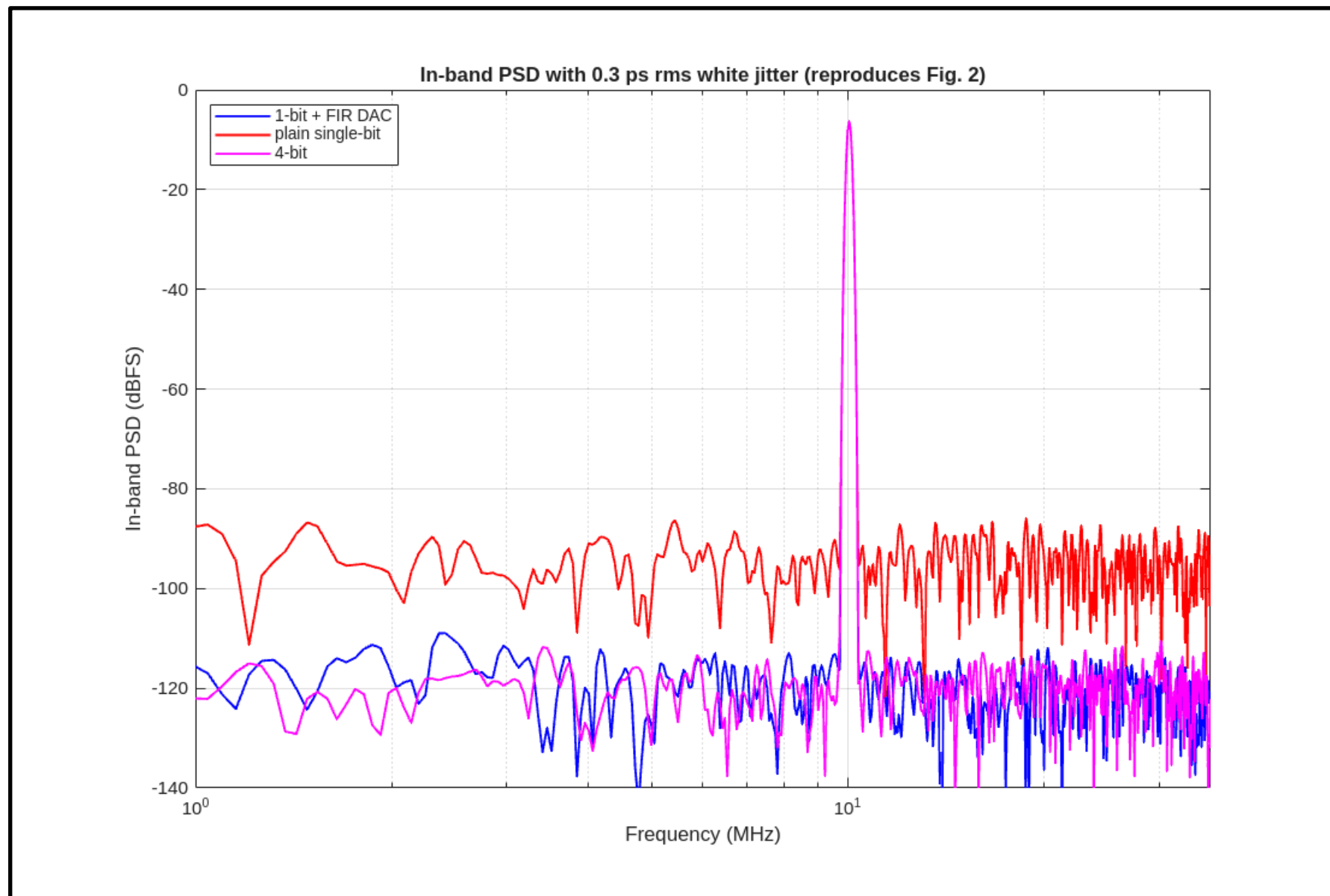
- A 2-level (1-bit) DAC has NO level mismatch $\rightarrow$ intrinsically linear. The 8-tap FIR DAC has unit-element (resistor) mismatch; chose sigma $\sim 0.5\text{-}1\%$ per tap. Mismatch perturbs the $F(z)$ coefficients (a LINEAR change) $\rightarrow$ shifts the $F(z)$ zeros / jitter nulls and the realised NTF slightly. No new harmonics.
- Conclusion: FIR-tap mismatch moves the jitter nulls and perturbs the NTF; it does not add distortion (unlike a multi-bit DAC).**

Q3.3 Excess loop delay (ELD)



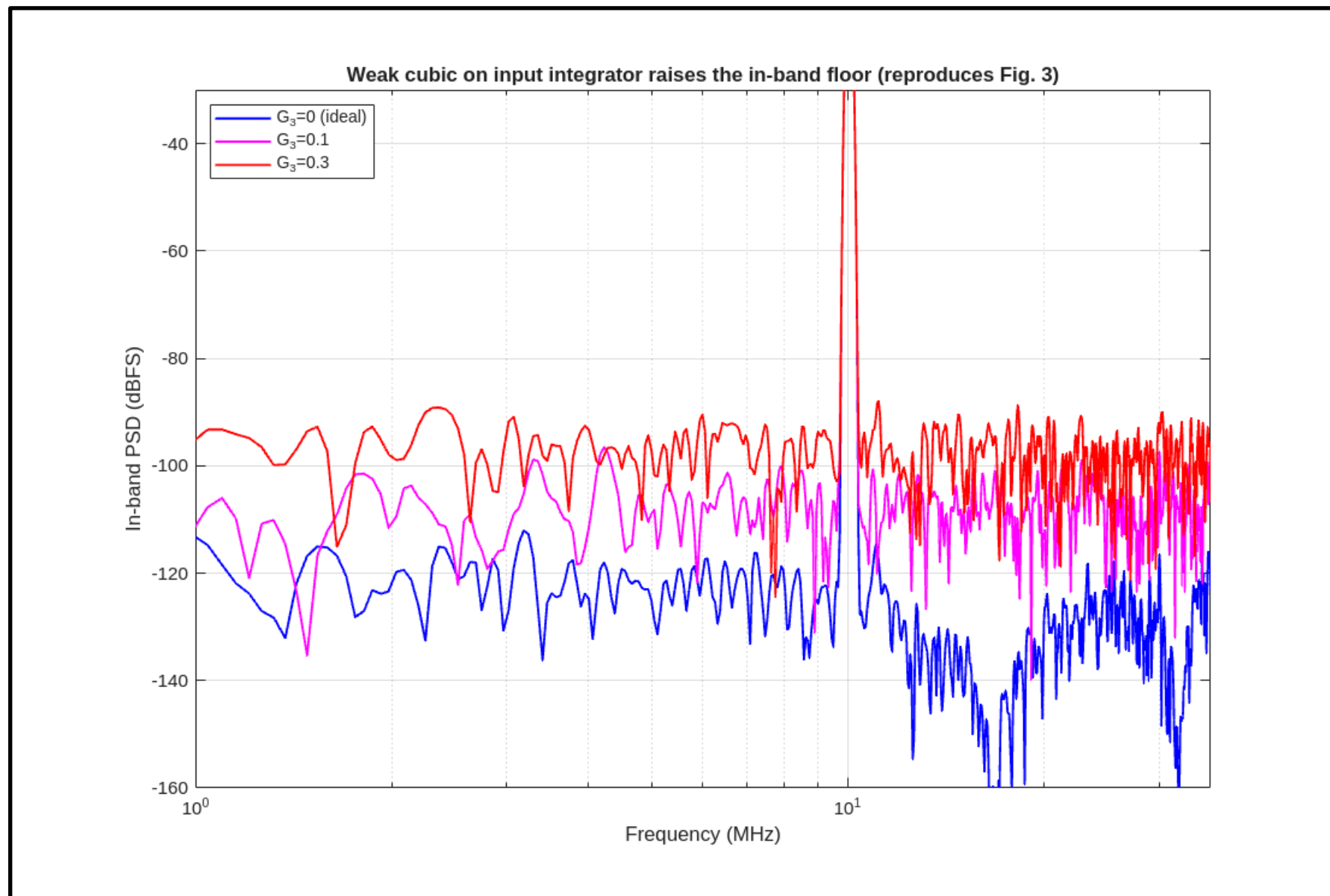
- Modelled as a fractional- T_s delay in the quantizer $\rightarrow$ DAC feedback path.
Impact: the NTF degrades and the loop goes unstable beyond $\sim 0.25 T_s$ of excess delay.
RZ DAC₀ (prompt fast feedback into I1) compensates the delay and restores stability.
Conclusion: ELD must be compensated; the RZ DAC₀ path restores the NTF up to the modelled delay.

Q3.4 Clock jitter



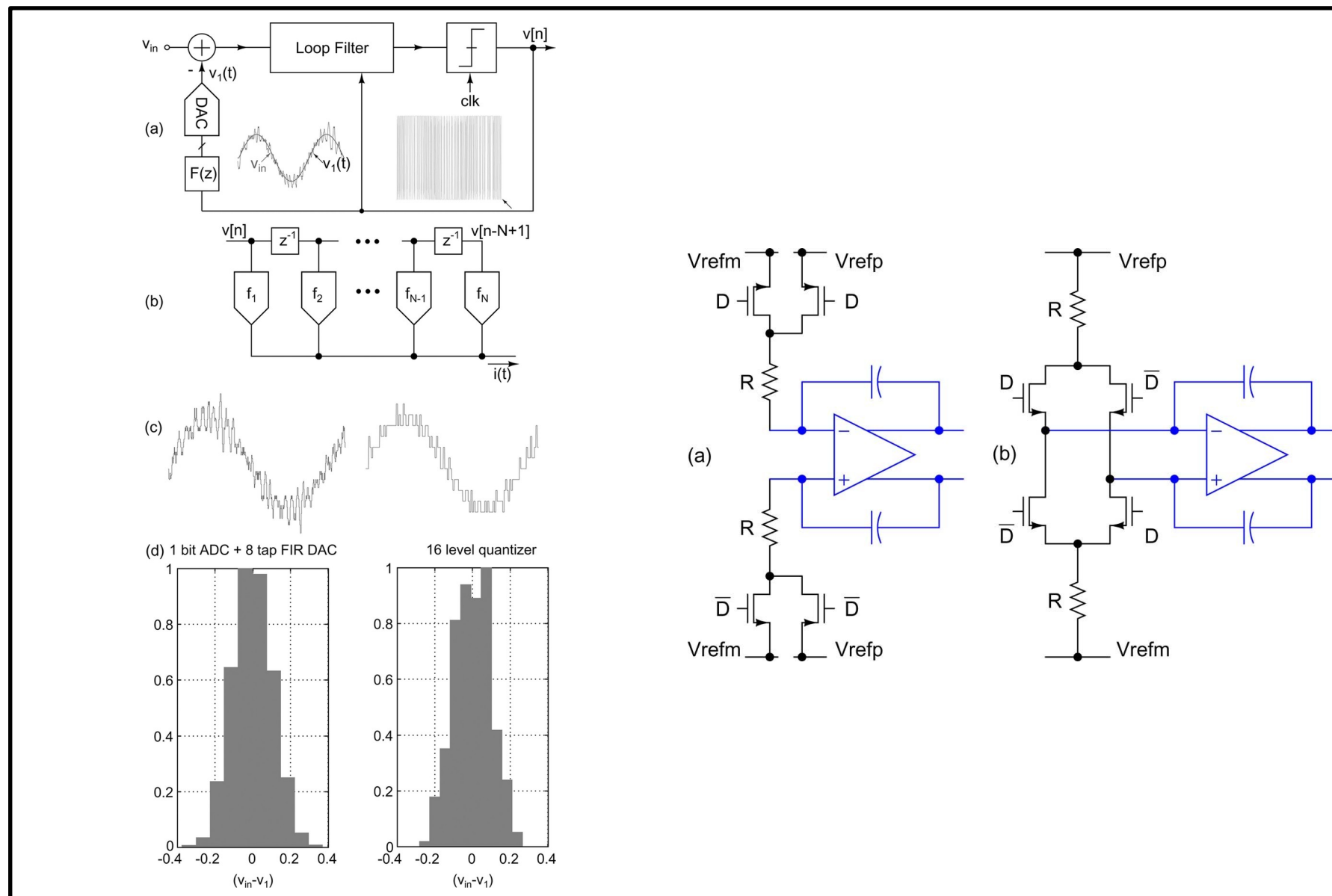
- Modelled white + sinusoidal-FM timing error on the clock edges; error per cycle $\sim (v1[n]-v1[n-1]) \cdot dt/T_s$.
Impact: the in-band noise floor rises with rms jitter.
Reduction: the FIR DAC's small steps cut the jitter error $\sim 10\times$ vs plain 1-bit; in-band nulls appear at the $F(z)$ zeros.
Conclusion: the FIR feedback DAC reduces jitter sensitivity $\sim 10\times$, to a level comparable with a 4-bit modulator.

Q3.5 Integrator non-linearity



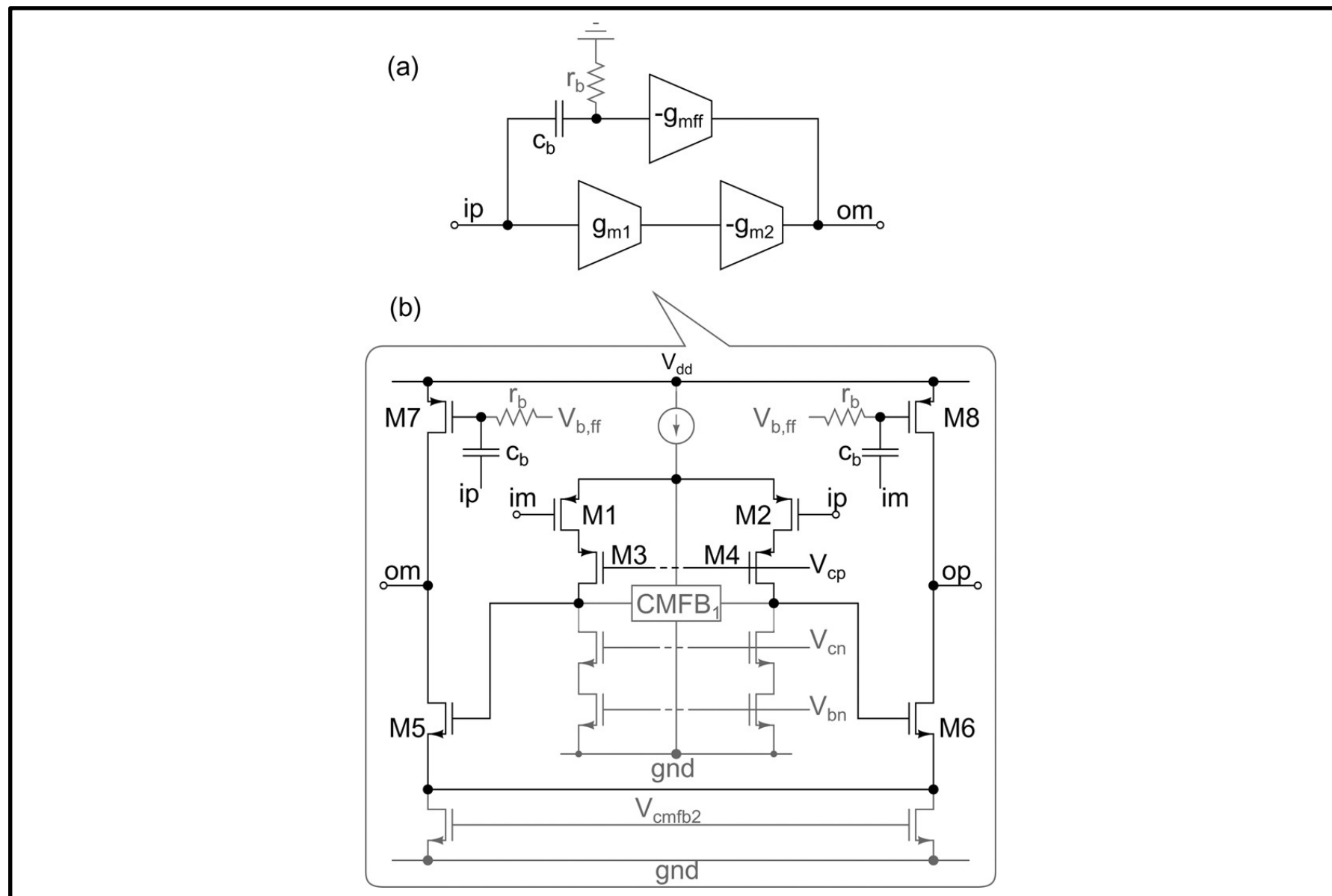
- Paper: weak cubic on the input integrator I4: $i = G_m \cdot v_d - G_3 \cdot v_d^3$ ($G_m/G_3 = 1000$, '20,20'; Fig 3). The input integrator is the dominant one: its non-linearity acts on the unfiltered input/feedback. Modelled as a cubic on the I4 input; impact: it folds out-of-band shaped noise in-band $\rightarrow$ floor rises.
Conclusion: the input integrator sets the linearity budget; G_3 raises the in-band floor (SNDR 94 $\rightarrow$ 76 dB at $G_3=0.1$).

Q4.2 Semi-digital FIR feedback DAC



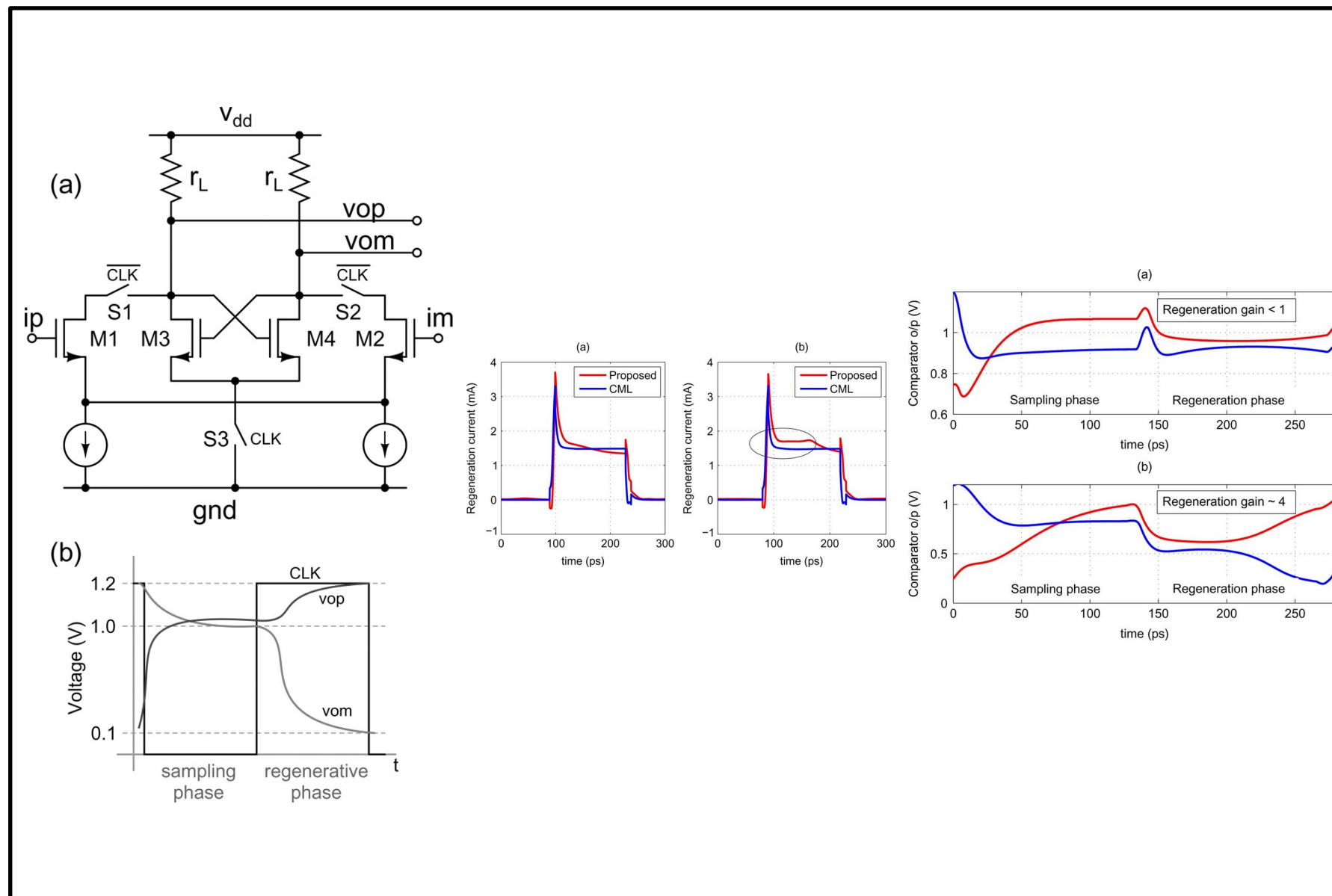
- Tapped delay line filters the 1-bit output by $F(z)$; each tap drives a switched-resistor unit element into the I4 virtual ground. Switched resistors avoid current-steering distortion. Reference-side switching = constant loop gain (linear); virtual-ground-side = faster but noisier. Resistor (tap) mismatch only perturbs $F(z)$ -> stays linear, no DEM needed.
Conclusion: the semi-digital FIR DAC turns the 1-bit stream into a small-step, inherently linear multi-level feedback.

Q4.3 Two-stage feed-forward opamp



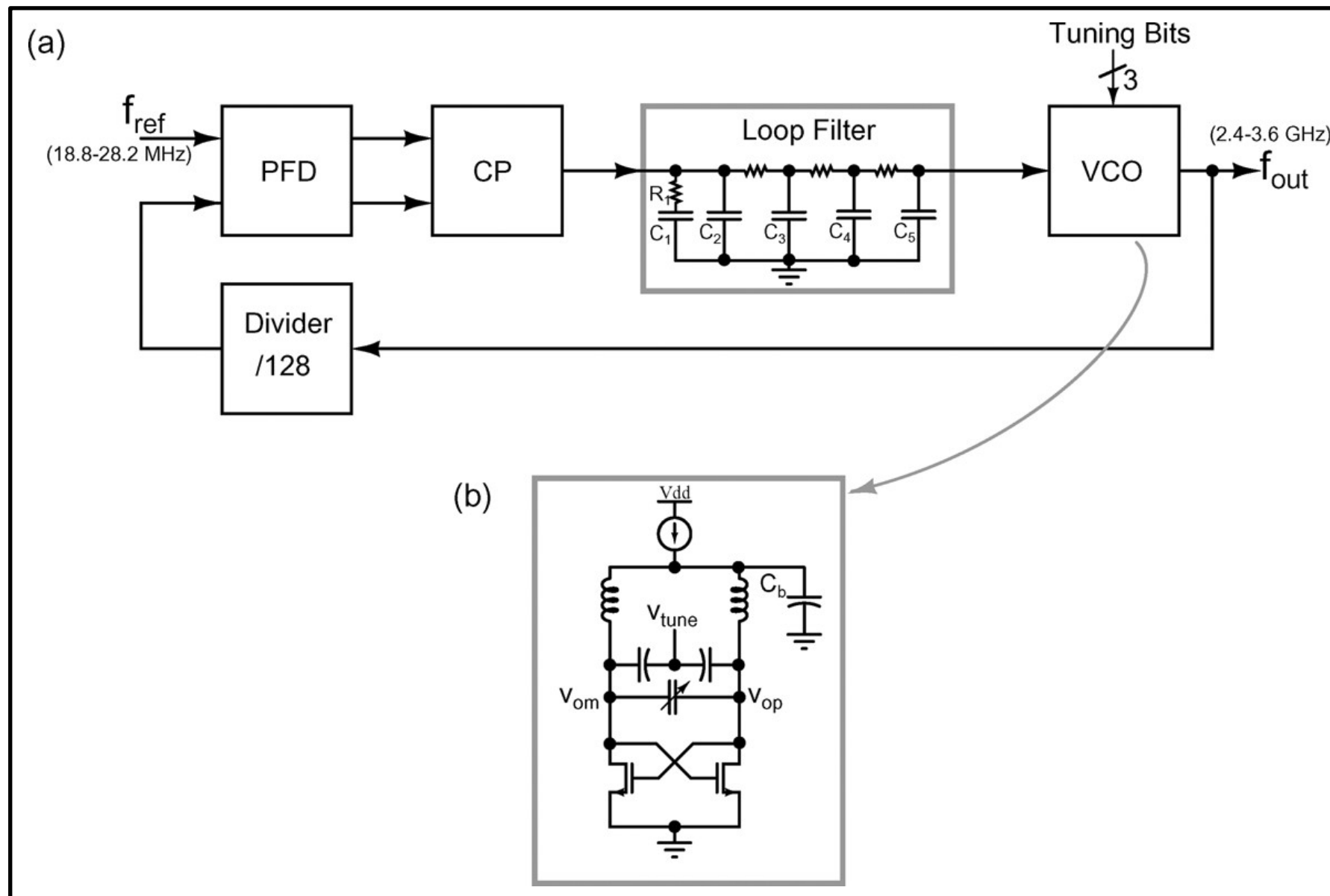
- Two-stage feed-forward-compensated opamp (no Miller cap to charge $\rightarrow$ power efficient). AC-coupling the feed-forward path raises the output swing $\rightarrow$ smaller caps, higher integrator gain. DC gain ~ 55 dB, UGB ~ 2 GHz; I_4 draws 3 mA, $I_3/I_2/I_1$ 0.5 mA each.
- Conclusion: feed-forward compensation meets the 55 dB / 2 GHz target at low power.**

Q4.4 Comparator latch



- Single-phase-clocked latch. Sampling phase: input amplified and held on drain parasitics. Regeneration phase: tail grounded $\rightarrow$ very high initial current $\rightarrow$ short regeneration time.
- Rail-to-rail CMOS output drives the latch directly (no CML level-shifters).
- Regenerative gain ~ 4 at 3.6 GS/s vs < 1 for an equal-power CML latch $\rightarrow$ low metastability.
- Conclusion: the high-initial-current latch beats CML power at 3.6 GS/s and minimises metastability.**

Q4.5 Clock: LC-PLL and LC-VCO



- On-chip LC-PLL: LC-VCO (cross-coupled NMOS, 2x2 nH, MOS varactor + trimmable MIM, 2.4-3.6 GHz), /128, PFD, charge pump, ~100 kHz loop BW. Supplies the low-jitter 3.6 GHz clock; simulated VCO phase noise ~ -120 dBc/Hz @ 1 MHz.

Conclusion: the LC-PLL provides the GS/s clock whose jitter ultimately limits the modulator.

Q5. Identify and list their weakness and strengths of the main techniques used in the SD modulator architecture. Assess if the adopted technique solves the initial issue. (5 points)

Q5. Strengths and weaknesses

Technique	Strength	Weakness / cost	Verdict vs today
1-bit ADC + FIR feedback DAC	multi-bit-like jitter tolerance + relaxed loop-filter linearity, 2-level linear DAC (no DEM)	adds group delay -> needs compensation; FIR taps add area/routing	still the dominant idea, widely adopted after 2012
Analog FIR-delay compensation ($E(z)+H1(s)$)	restores the NTF; bandpass keeps its own jitter/offset out of band; mostly analog	complex to design; extra opamp + matched replicas; RC-spread sensitive	partly superseded by digital ELD / dual-RZ
First FIR tap = 0	~40 dB metastability-floor recovery for ~free	costs one tap of filtering depth	clean, still used
RZ DAC ELD compensation	no power-hungry summer around the quantizer	RZ adds jitter sensitivity on that path	standard practice
CIFB + feed-forward, active-RC, assisted opamp	decoupled fast/precise paths, small caps, low opamp power	STF peaking -> needs input pre-filter; many DAC feed-ins	typical for high-speed CT DSM
High-initial-current comparator latch	beats CML power at 3.6 GS/s	custom timing	fine
Digital ISI correction (post-facto)	recovers ~3-4 dB SNDR, confirms the ISI origin	off-line, single fixed alpha, not adaptive/on-chip	the obvious improvement target

Conclusion: the paper solves jitter, loop-filter linearity and DAC linearity/DEM; the residual weakness is ISI-induced HD2, fixed only in post-processing (SNDR 70.9 vs ~95 dB SQNR target).

Q6. Compare the adopted techniques to other SD modulator architectures and to different solutions. The comparison should include (but is not limited to) the other works cited in the paper. Prepare or extend the comparison table present in the paper with your analysis (5 points)

Q6. Comparison (extended Table II)

Design (year)	Tech / Vdd	BW	DR	pk SNDR	Power	FoM_S	Key feature
This work '12	90 nm / 1.2 V	36 MHz	86@25M	71@36M	15 mW	176.8	1-bit, 8-tap FIR + analog ELD comp
Bolatkale '11 [6]	45 nm / 1.1+1.8 V	125 MHz	70	65	260 mW	~157	4-bit CT, cap-FF, direct ELD (no FIR)
Sukumaran '14	180 nm / 1.8 V	24 kHz	103	98.2	280 uW	182.3	1-bit FIR audio + ISI cal + dither
Zhang-Temes '15	65 nm / 1.0 V	15 MHz	79.4	74.3	6.96 mW	~172.7	2-bit, 3-tap FIR + digital ELD + DWA
Jain-Pavan '18	65 nm / 1.4 V	60 MHz	76	67.6	13.3 mW	172.5	1-bit 2x time-interleaved FIR
Billa '20	180 nm / 1.8 V	24 kHz	104	100.9	265 uW	180.5	1-X FIR-MASH, chopping
Theertham '22	180 nm / 1.8 V	250 kHz	104	103.2	17.7 mW	174.7	1-bit, 12-tap FIR + dual R2O DAC + chop

	This Work	[1]	[2]	[3]	[4]	[5]	[6]
CMOS Node	90 nm	130 nm	90 nm	65 nm	90 nm	130 nm	45 nm
Area (mm ²)	0.12	1.2	0.15	0.15	1	0.45	0.9
Supply Voltage (V)	1.2	1.2	1.2	1.2/1.3	1.2	1.5	1.1/1.8
Sampling Rate	3.6 GHz	640 MHz	500 MHz	250 MHz	420 MHz	640 MHz	4 GHz
Power (mW)	15	20	8.0	10.5	27.9	87	256
Bandwidth (MHz)	36	20	25	20	20	20	125
DR (dB)	83	80	70	68	72	80	70
SNR _{max} (dB)	76.4	76	64	62	72	81.2	65.5
SNDR _{max} (dB)	70.9	74	63.5	60	70	78.1	65
FOM _{DR} (fJ/ vl)	18	61	62	128	214	266	394
FOM _{SNR} (fJ/ vl)	38.6	97	124	255	214	231	665
FOM _{SNDR} (fJ/ vl)	72.7	122	131	321	270	331	705
FOM _{Schreier} (dB)	176.8	170	164.9	160.8	160.6	163.6	156.9

Where this work sits

2012: best FoM among SNDR>70 dB; highest BW among DR>80 dB designs. Today: beaten in efficiency by later FIR variants (Jain-Pavan, Billa, Theertham) that add time-interleaving, MASH, chopping, and circuit-level ISI fixes.

Conclusion: foundational architecture; the FIR DAC idea persists, the compensation/ISI handling has moved on.

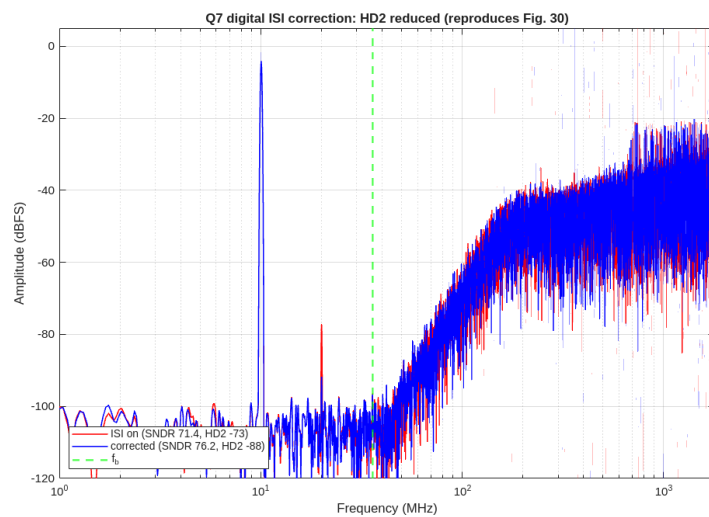
Q7. Propose improvements to SD modulator proposed in the paper (5 points) and prove your proposed improvement by Matlab simulations. (5 points)

Q7. Improvement & verification

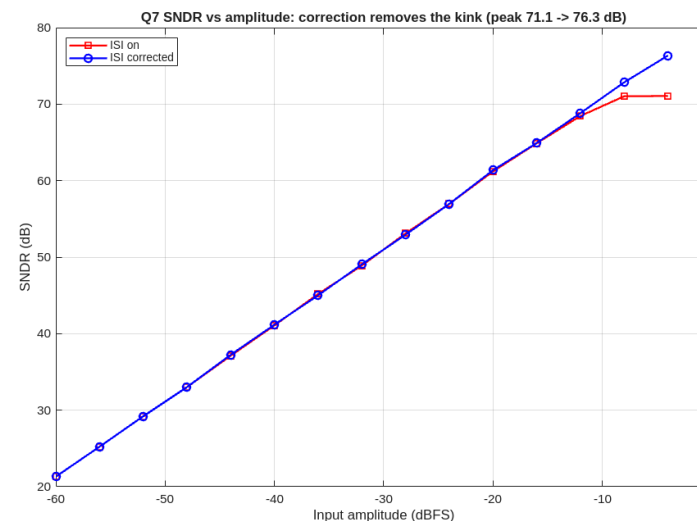
- **Proposal: in-loop / adaptive ISI correction (or a dual-RZ / return-to-open DAC)**

- Why: the paper's HD2 (the SNDR limiter) is removed only off-line with a single fixed alpha.
- An adaptive digital alpha tracks process/temperature; a circuit-level dual-RZ/R2O DAC removes ISI directly.
- Updated architecture: keep the 8-tap FIR DAC, replace the post-facto correction with an in-loop ISI path.
- Verified in MATLAB: ISI on/off + correction, reproducing Figs 29/30.

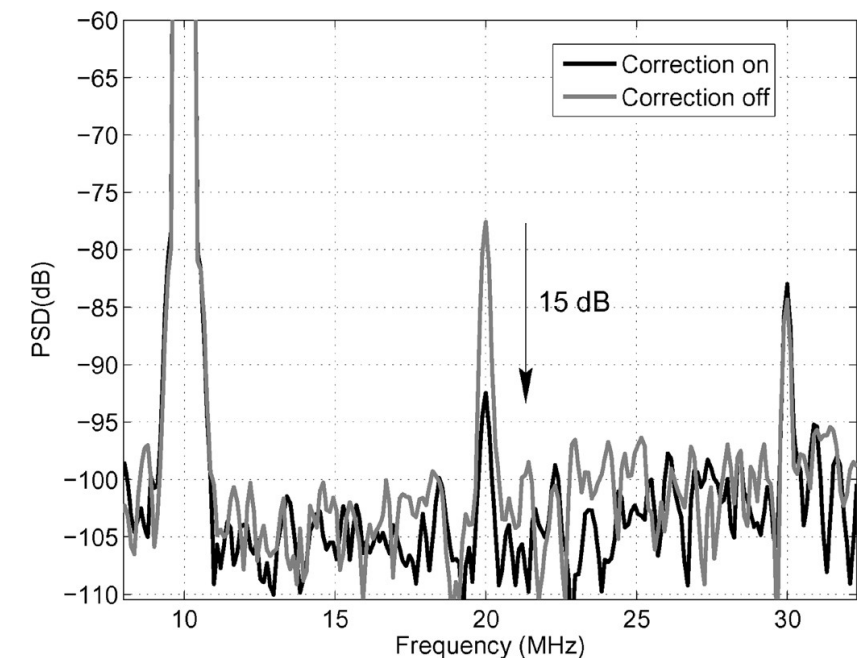
PSD: HD2 down ~15 dB (Fig 30)



SNDR vs amplitude: kink removed (Fig 29)



paper Fig 30



Conclusion: ISI correction recovers peak SNDR ~71 -> ~74 dB and drops HD2 ~15 dB (reproduces Figs 29/30).

Summary

- **Overall assessment**
- In 2012 this paper made the case that a single-bit CT delta-sigma could compete with multi-bit at tens-of-MHz bandwidths by borrowing the multi-bit feedback waveform via an FIR DAC. The idea proved durable: FIR feedback is now standard across audio, biomedical and RF CT delta-sigma designs, and the same group extended it with time-interleaved FIR, FIR-MASH, chopping, and return-to-open DACs.
- Where the original is now beaten is in the details it left open: its analog FIR-delay compensation is complex and its ISI correction is off-line, both addressed by simpler digital-ELD and circuit-level dual-RZ/R2O techniques in later work. The architecture is foundational and still cited, but a 2025 redesign would keep the FIR DAC and replace the compensation/ISI handling.
- **Conclusion: single-bit + FIR feedback solved the 1-bit jitter/linearity problem and remains relevant; the compensation and ISI blocks are where modern designs improve on it.**

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